



DualCard User Manual

DUALi Inc.

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Our product development center is located in South Korea to provide technical support. For assistance, contact our technical support team below.

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Revision History

- 2009.02. (Ver. 1.0) : First Release
- 2009.05. (Ver. 1.01) : 15693 (Optional. DE-620R) Function Added
- 2010.04. (Ver. 1.02) : Mifare plus function tap added
- 2013.08. (Ver. 1.1) : Layout composition modified (DualCard version 3.1)
 - Update Script input more user-friendly
 - Mifare tab: 4/7/10 bytes UID support function added
 - NFC tab : LLCP/SNEP (ISO 18092) and NFC Barcode test function added.
 - UTIL tab : device number check, serial communication speed select function, Triple DES function added.
- 2015.03.11 (Ver.2.0) : DualCard layout composition modified. (DualCard version 4.0)
 - DESFire function improved. (Batch function)
 - Util tab revision (Util 1/ Util 2)
 - PCSC mode test function (all tabs)
- 2016.04.12 (Ver.2.1) : MIFARE Ultralight C tap added (Ch.4.4)
 - Contents modified
 - Felica, DesFire features improved.
- 2018.07.31 (Ver.2.2): Layout Update

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Introduction

A. Intended Readership

This manual was written while keeping in mind the end-user. This manual is intended to act as a reference.

B. Purpose

This manual describes the various functions that will be provided in the application. The purpose of this document is to give detailed descriptions about the Application Controls and the Interfaces with readers and cards.

C. Scope

The DualCard program supports the following DUALi readers, which support the following tags and cards.

Tags and Cards

- ISO 7816
- ISO 14443 Part 3 and Part 4 (Type A and Type B)
- ISO 15693
- NFC
- Mifare Series (registered trademarks of NXP Semiconductors)
- DesFire (registered trademarks of NXP Semiconductors)
- Felica (registered trademark of SONY Corporation)

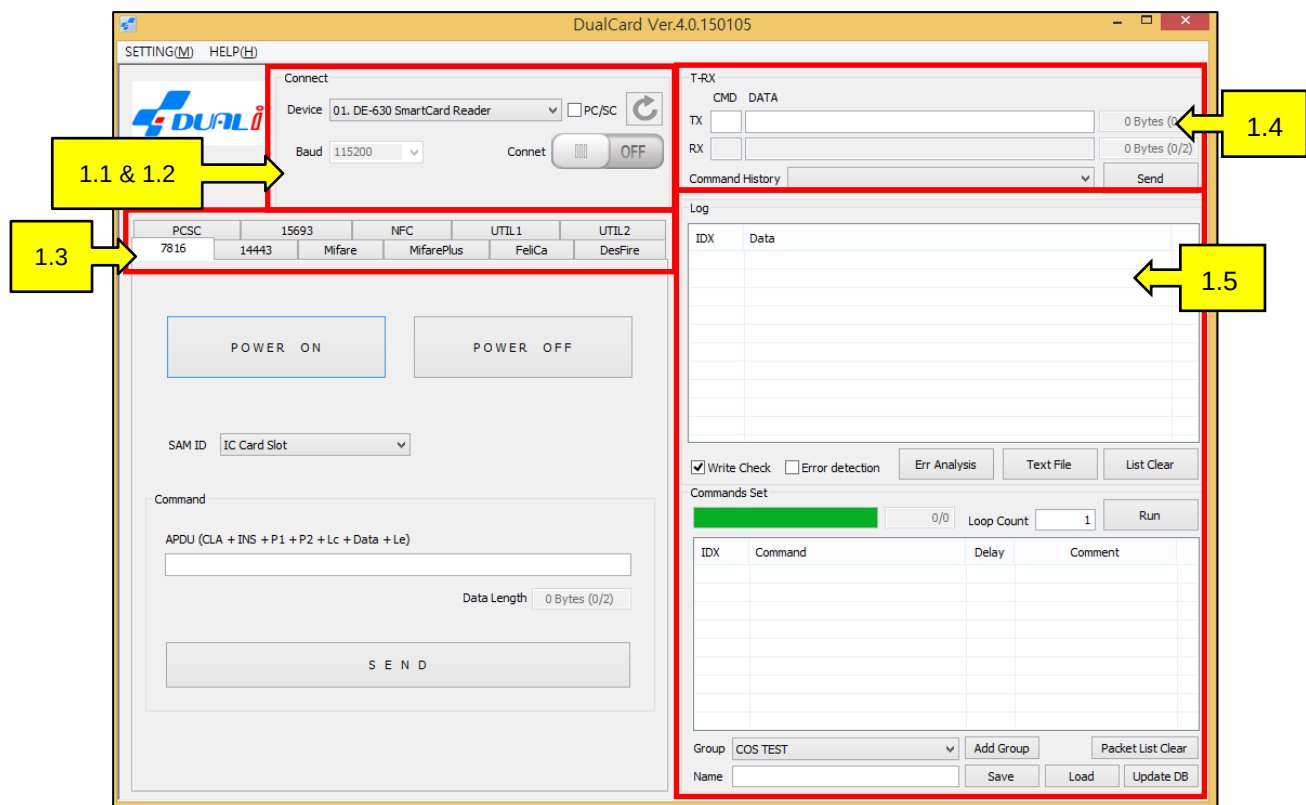
DUALi Readers

- DE-620/620R/620V/620A, DE-630/631, Dragon, DE-930/950 FDK, NFC PAD
- All DUALi modules

Supported Operating Systems:

Windows 2000, Windows XP, Windows Vista (32bit), Windows 7 (32/64bit), Windows 8, Windows 8.1, Windows 10

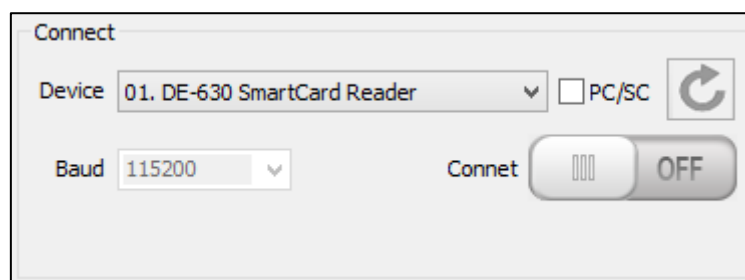
1. DualCard Layout



< Figure 1-1 >

DualCard Main Layout

1.1 Connect (Vender Mode)



<Figure 1-5>

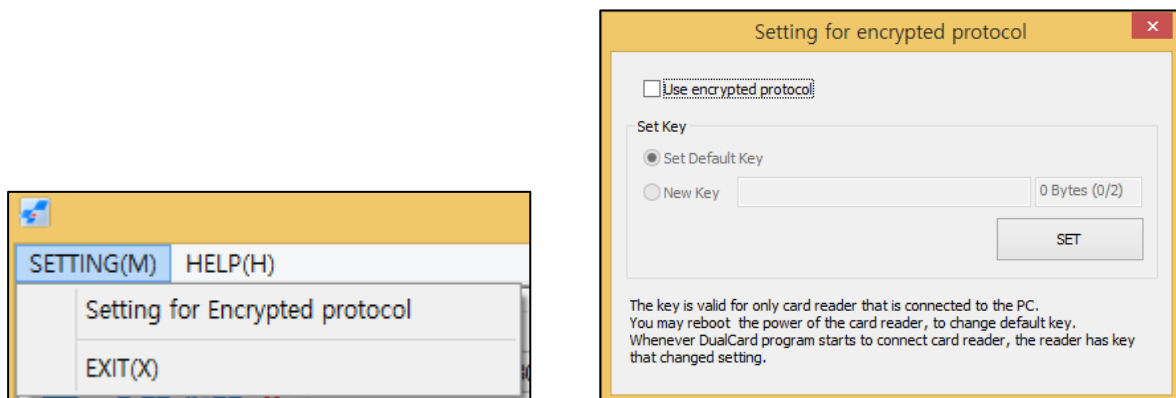
Connect, Vendor Mode

Once a device is connected, the product name is displayed on the “Device” dropdown box. The support interface can either be USB or Hyper Terminal (RS-232). If UART is used, the standard communication speed will be set to 115200bps.

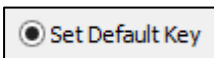
- : Click to start a test of the reader, once a reader is selected from the “Device” list.
- : Click to refresh the list of devices when an additional reader is connected while a test is running.
- : Status of a connection changes to “ON” when a device is connected
DualCardDll API: DE_InitPort([PORT],[BAUD]);
- : Status of a connection changes to “OFF” when a device is disconnected
DualCardDll API: DE_ClosePort(m_pDoc->m_nPort);

You are able to use encrypted protocols between a PC and a DUALi device with the DualCard Version 4.0. To use this function, select “Setting for Encrypted protocol” from the main menu._

*To use this function, you must update the firmware. Contact the DUALi Lab for assistance.

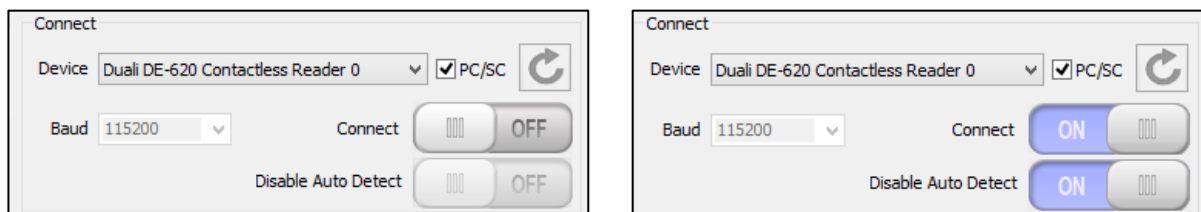


<Figure 1-3>
Encrypted Protocol

- : Check the box to activate encrypted communication
- : Select this option to use the default key in the terminal for encrypted communications. The reader is set to the default key after it has been reset.

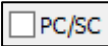
- : Select this option to use the User's key for encrypted communication. The new key setting will be valid after exiting and reopening the DualCard system. However, after the reader has been reset, the key setting will return to the default key.
- : Click to save settings. While this function is activated, you can find the  under the DUALi logo in program in the main menu, <Figure 1-1>.

1.2 Connect (PC/SC mode)



<Figure 1-4>

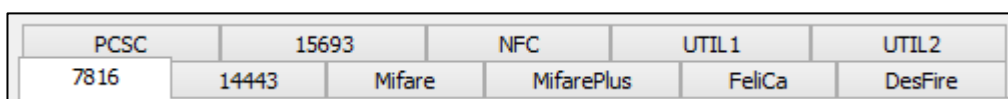
Connect PC/SC Mode

- : Check the PC/SC box to have PC/SC devices populated in the device list
- : Click once you have placed the RF card on the reader
- : Click to "OFF" to disable the auto detect function. This allows the use of all function tabs in DualCard.

DualCardDII API :

```
DE_SCardConnect([PORT]);
DE_SCardDisconnect(m_pDoc->m_nPort);
```

1.3 Function Tabs



<Figure 1-5>

Function Tabs

- The figure above shows the various protocols offered in the system.
- DualCard supports ISO 7816, ISO14443 A/B, Mifare, MifarePlus, FeliCa, DesFire, PCSC, ISO15693, NFC and Utility functions. Please double check the protocols that the device supports.

1.4 Command Input

T-RX

CMD DATA

TX 0 Bytes (0/2)

RX 0 Bytes (0/2)

Command History

<Figure 1-6>
T-RX

- The T-RX group box, <Figure 1-6>, is for the user to input direct commands from the protocol specifications. Each command must be no more than 1 byte size (as Hexadecimal).

	CMD	DATA
TX		

- : Input the command (2 digit Hexadecimal) in the first field and the data in the second field. Then click "Send".

Send

- : Click to execute a command

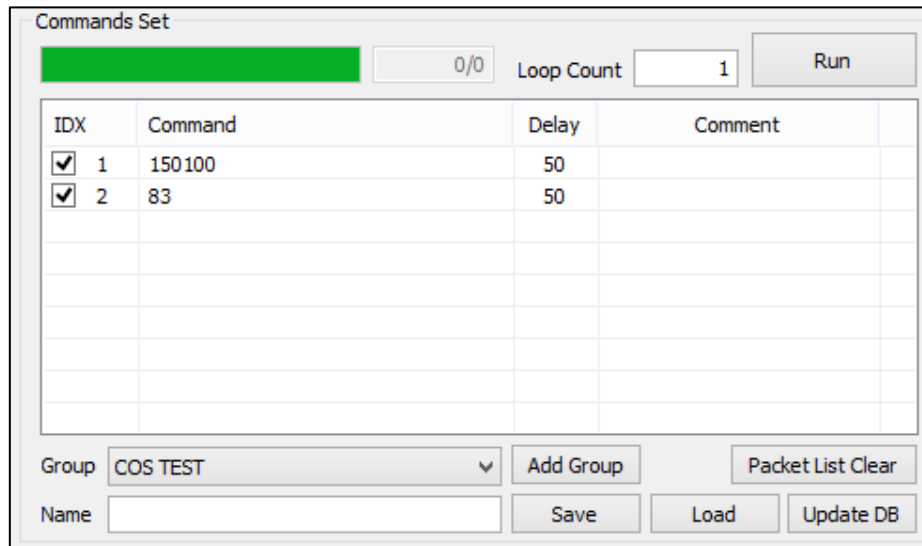
DualCardDll API: DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT])

RX

-  : The response value from reader/card is displayed here. The response code is displayed in the first field and the response data in the second field.

Command History		Send
	3A0100 3A02000-45A000001 4C00000 100 150 100 83	

- : The Dropbox displays the 10 most recent commands in order.



Commands Set

0/0 Loop Count 1 Run

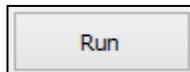
IDX	Command	Delay	Comment
☒ 1	150100	50	
☒ 2	83	50	

Group COS TEST Add Group Packet List Clear

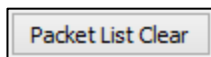
Name Save Load Update DB

<Figure 1-10>
Packet List

- Double click the commands from the Log, <Figure 1-7>. The selected commands will be added to the packet list, <Figure 1-10>.

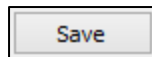


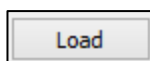
- : Click to execute commands, the commands with the checked boxes will be executed the number of times that is in the "Loop Count" field, .



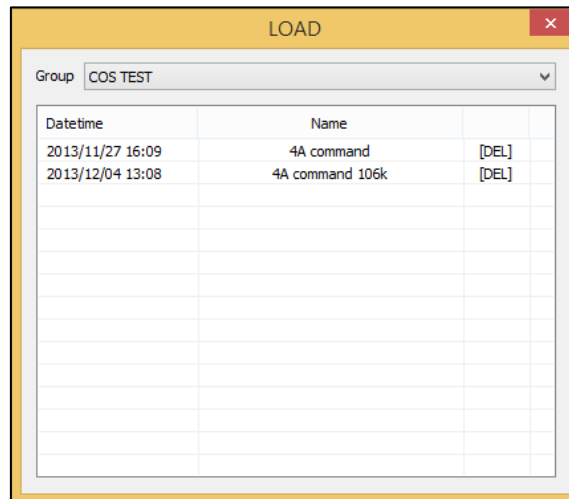
- : Click to clear the Packet List.

Name

- : input a file name and click  to save the full command list as a file. It will be saved to the packet list in the database.



- : Click to display a saved file to the list. The selected file will be displayed like in <Figure 1-11>.



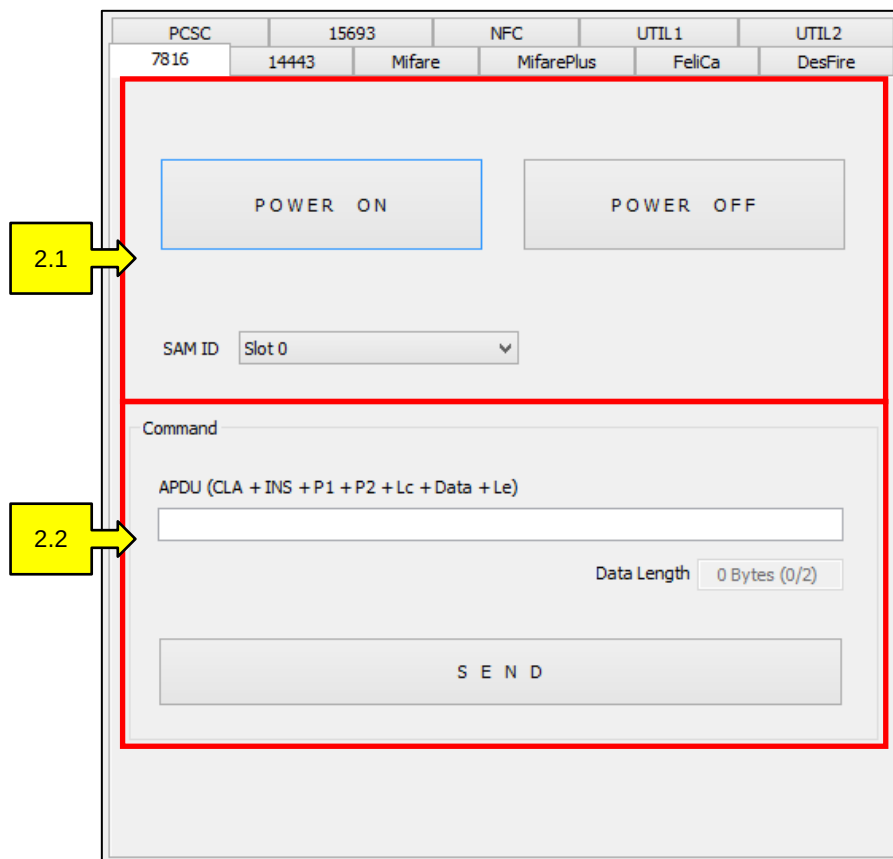
<Figure 1-11>

Load

- From <Figure 1-11>, double click on a command name in the “Name” field to add command sets to the packet list, <Figure 1-10>.
- From <Figure 1-11>, double click on “[DEL]”, next to the file name to delete a command set

2. 7816

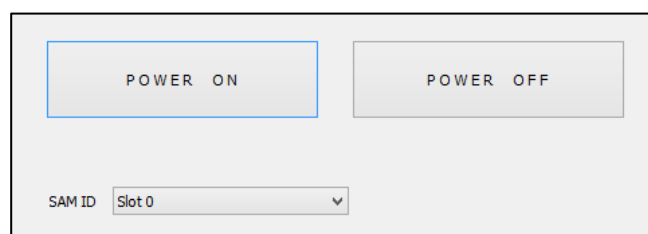
The 7816 tab, <Figure 2-1>, supports communication with ISO 7816 IC contact cards. From this tab, you can manage the Power On/Off function and command execution.



<Figure 2-1>

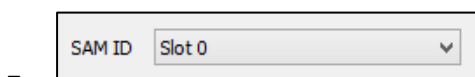
7816 Tab

2.1 Power On/Off



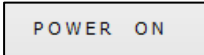
<Figure 2-2>

Power On/Off



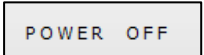
: Select the appropriate card slot to turn the power

on/off to.

- : Click to turn the power on for the card slot (SAM Slot/ IC Card Slot).

DualCardDII API : DE_IC_PowerOn([PORT], [SLOT NO], [RLEN], [RBUF]);

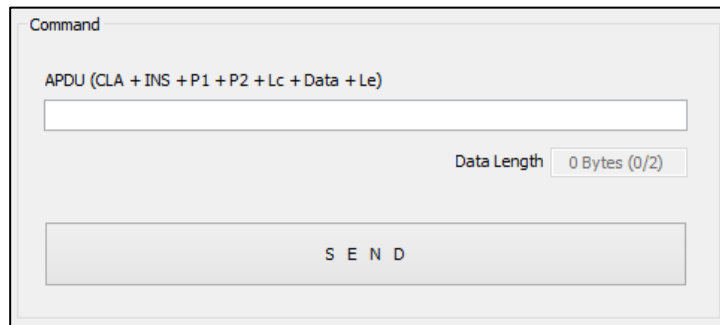
DUALi RW Protocol : C0[SLOT NO]00

- : Click to turn the power off for the card slot (SAM Slot/ IC Card Slot).

DualCardDII API : DE_IC_PowerOff([PORT], [SLOT NO]);

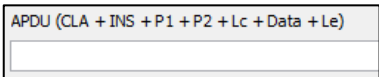
DUALi RW Protocol : C5[SLOT NO]

2.2 Command



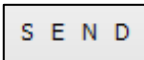
<Figure 2-3>

7816 Command

- : Input a command (digit Hexadecimal), that complies with ISO 7816, and click "Send". A response will be shown on the log, <Figure 1-7>.

DualCardDII API : DE_IC_Case4 ([PORT], [SLOTNUM], [SLEN] , [SBUF], [RLEN], [RBUF]);

DUALi RW Protocol : C4[SLOT NO][APDU]

- : Click to execute a command.

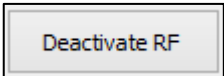
3. 14443

The 14443 tab, <Figure 3-1>, supports communication with ISO 14443 A/B compliant cards. In this tab, you can manage the Type A command set, Type B command set, card detect function, User command transmit, and RF field control. For the technical detail of the protocol, refer to ISO 14443 Specifications.

<Figure 3-1>

14443 Tab

3.1 RF field Activation

-  : Select to de-activate the RF field.

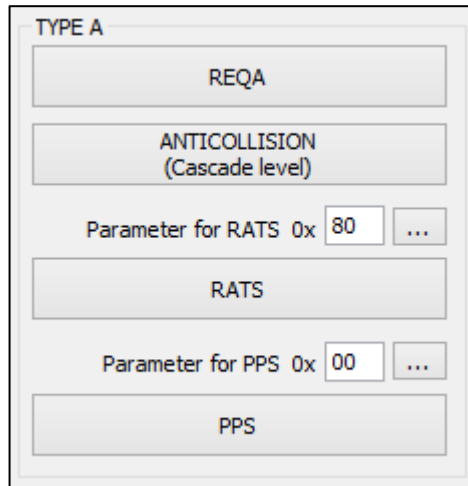
DualCardDll API : DE_RFOff([PORT]);

DUALi RW Protocol : 11

- Activate RF

 : select to activate the RF field.
 DualCardDII API : DE_RFOn([PORT]);
 DUALi RW Protocol : 10

3.2 Type A



<Figure 3-2>
Type A

- REQA

 : Click to execute the REQA command. A response will be shown on the log, <Figure 1-7>.
 DualCardDII API : DEA_Idle_Req([PORT],[RLEN],[RBUF]);
 DUALi RW Protocol : 21
- ANTICOLLISION
(Cascade level)

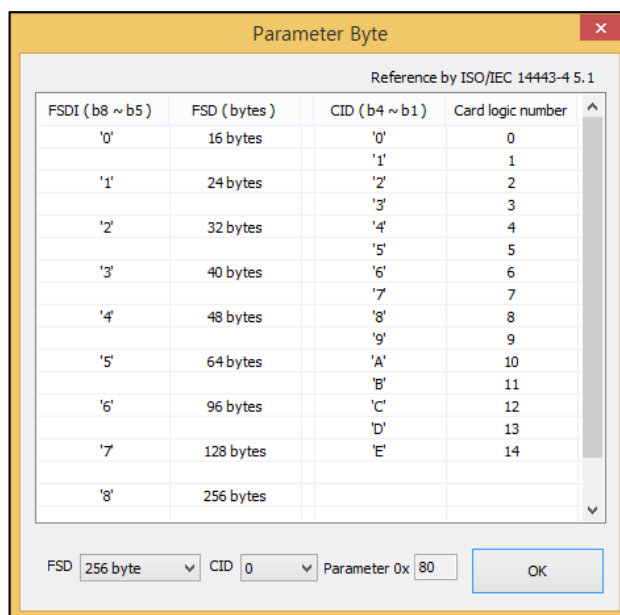
 : Click to execute and get respon”, which supports cascade level commands. A response will be shown on the log, <Figure 1-7>.
 DualCardDII API : DEA_AntiSelLevel([PORT],[RLEN],[RBUF]);
 DUALi RW Protocol : 3D
- RATS

 : Click to execute RATS commands. A response will be shown on the log, <Figure 1-7>.
 DualCardDII API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF]);
 DUALi RW Protocol : 41E0[RATS][TIMEOUT]
- Parameter for RATS 0x 80 ...

 : Input the parameter byte for RATS or click the “...” button to

automatically generate the value.

- : Click to display the parameter byte setting window for RATS, <Figure3-3>.



Parameter Byte

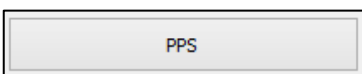
Reference by ISO/IEC 14443-4 5.1

FSDI (b8 ~ b5)	FSD (bytes)	CID (b4 ~ b1)	Card logic number
'0'	16 bytes	'0'	0
'1'	24 bytes	'1'	1
'2'	32 bytes	'2'	2
'3'	40 bytes	'3'	3
'4'	48 bytes	'4'	4
'5'	64 bytes	'5'	5
'6'	96 bytes	'6'	6
'7'	128 bytes	'7'	7
'8'	256 bytes	'8'	8
		'9'	9
		'A'	10
		'B'	11
		'C'	12
		'D'	13
		'E'	14

FSD: 256 byte CID: 0 Parameter 0x: 80 OK

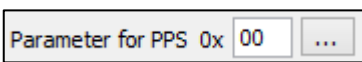
<Figure 3-3>

Parameter Byte, RATS

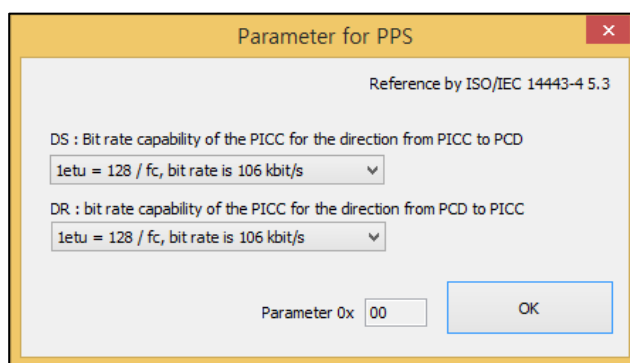
- : Click to execute the PPS Command. A response will be shown on the log, <Figure 1-7>.

DualCardDII API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol : 41[PPS]1100[TIMEOUT]

- : Input the parameter byte for PPS or click the “...” button to automatically generate the value.

- : Click to display the parameter byte setting window for PPS <Figure 3-4>.



Parameter for PPS

Reference by ISO/IEC 14443-4 5.3

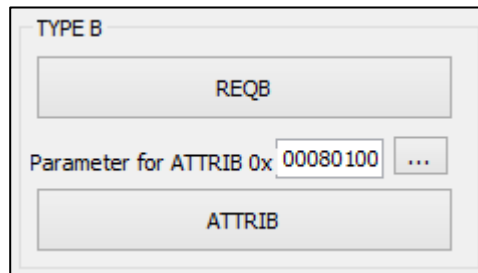
DS : Bit rate capability of the PICC for the direction from PICC to PCD
 1etu = 128 / fc, bit rate is 106 kbit/s

DR : bit rate capability of the PICC for the direction from PCD to PICC
 1etu = 128 / fc, bit rate is 106 kbit/s

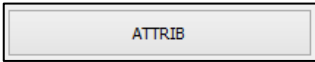
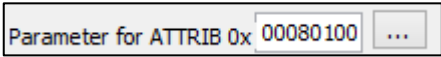
Parameter 0x: 00 OK

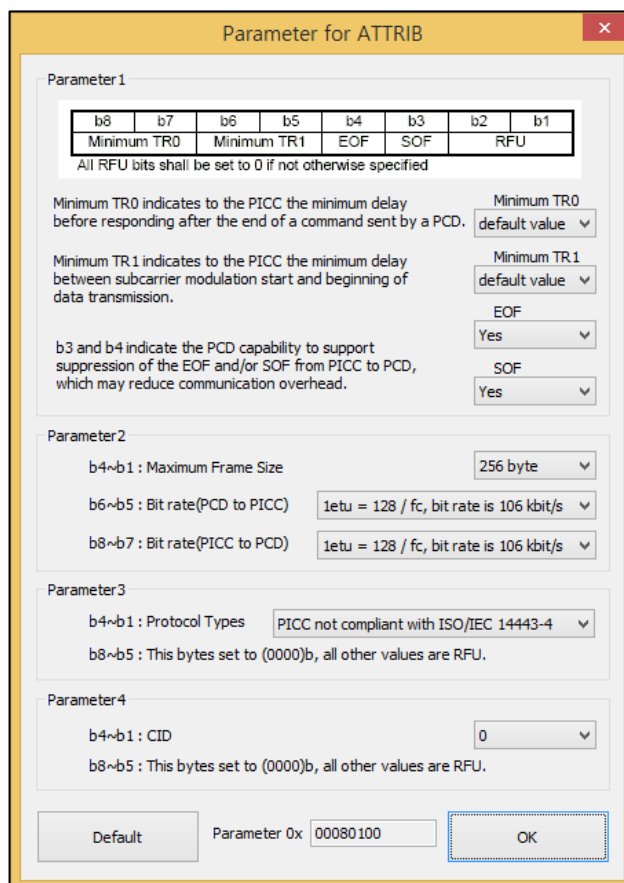
<Figure 3-4>
Parameter Byte, PPS

3.3 Type B



<Figure 3-5>
Type B

- : Click to execute the REQB command. A response will be displayed in the log <Figure 1-7>.
DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]);
DUALi RW Protocol : 6005000050
- : Click to execute the ATTRIB command. A response will be displayed in the log <Figure 1-7>.
DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]);
DUALi RW Protocol : 601D[ID][PARAMETER][TIMEOUT]
- : Input the parameter byte for ATTRIB or click the “...” button to automatically generate the value.
- : Click to display the parameter byte setting window for ATTRIB, <Figure 3-6>.



Parameter for ATTRIB

Parameter1

b8	b7	b6	b5	b4	b3	b2	b1
Minimum TR0	Minimum TR1	EOF	SOF	RFU			

All RFU bits shall be set to 0 if not otherwise specified

Minimum TR0 indicates to the PICC the minimum delay before responding after the end of a command sent by a PCD. Minimum TR0 default value

Minimum TR1 indicates to the PICC the minimum delay between subcarrier modulation start and beginning of data transmission. Minimum TR1 default value

b3 and b4 indicate the PCD capability to support suppression of the EOF and/or SOF from PICC to PCD, which may reduce communication overhead. EOF Yes SOF Yes

Parameter2

b4~b1 : Maximum Frame Size 256 byte

b6~b5 : Bit rate(PCD to PICC) 1etu = 128 / fc, bit rate is 106 kbit/s

b8~b7 : Bit rate(PICC to PCD) 1etu = 128 / fc, bit rate is 106 kbit/s

Parameter3

b4~b1 : Protocol Types PICC not compliant with ISO/IEC 14443-4

b8~b5 : This bytes set to (0000)b, all other values are RFU.

Parameter4

b4~b1 : CID 0

b8~b5 : This bytes set to (0000)b, all other values are RFU.

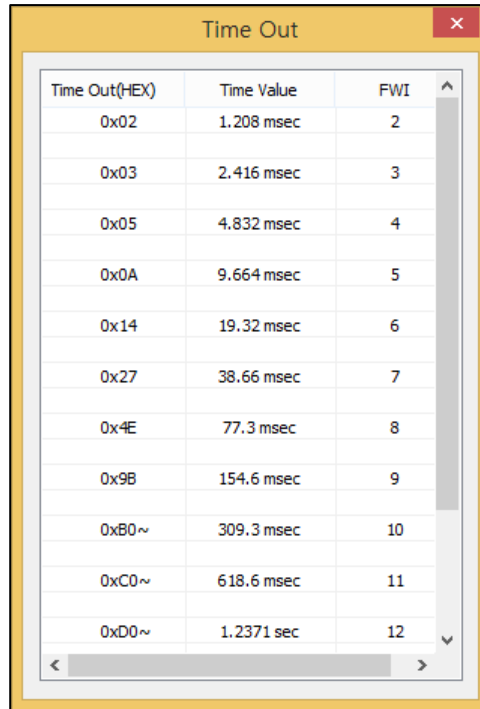
Default Parameter 0x 00080100 OK

<Figure 3-6>

Parameter Byte, ATTRIB

3.4 Common Function

- Time Out 0x 50 ...: Input the number of times it takes for the device to send commands and get a response or click the “...” button to automatically generate the value.
- ...: Click to display the “Time-Out” window, <Figure 3-7>.
- Transparent: Click to transmit a specific protocol or APDU type command that complies with ISO 14443 specifications. Input the PCB, CID and NAD according to your card specification.



Time Out(HEX)	Time Value	FWI
0x02	1.208 msec	2
0x03	2.416 msec	3
0x05	4.832 msec	4
0x0A	9.664 msec	5
0x14	19.32 msec	6
0x27	38.66 msec	7
0x4E	77.3 msec	8
0x9B	154.6 msec	9
0xB0~	309.3 msec	10
0xC0~	618.6 msec	11
0xD0~	1.2371 sec	12

<Figure 3-7>

Time-Out Setting

DualCardDII API :

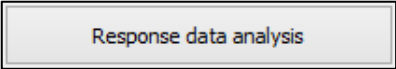
Type A : DEA_Transparent([PORT],[DATA LEN],[DATA],[TIMEOUT],[RLEN],[RBUF]);

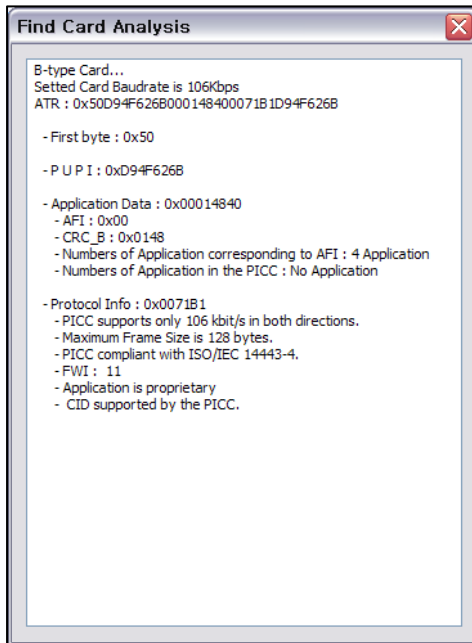
Type B : DEB_Transparent([PORT],[DATA LEN],[DATA],[TIMEOUT],[RLEN],[RBUF]);

DUALi RW Protocol :

Type A : 41[DATA][TIMEOUT]

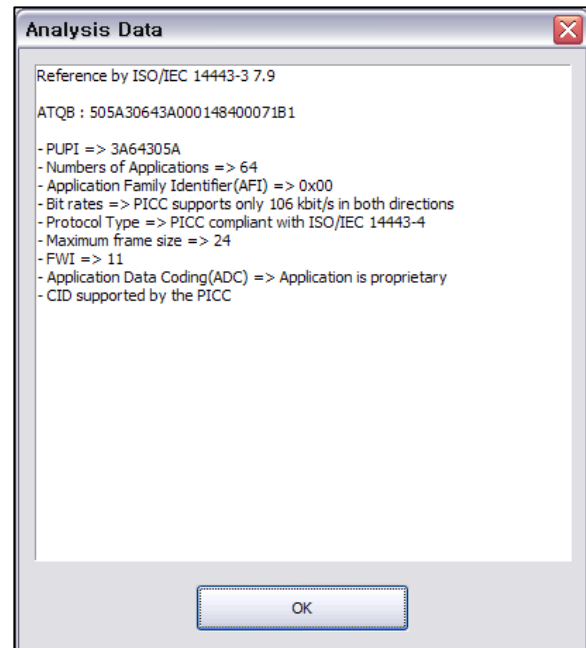
Type B : 60[DATA][TIMEOUT]

- : Click after executing the [REQA], [Anti-collision], [Select], [RATS], [REQB], or [Detect Card] commands. The response data analysis window will be displayed, as shown in <Figure 3-8> and <Figure 3-9>.



<Figure 3-8>

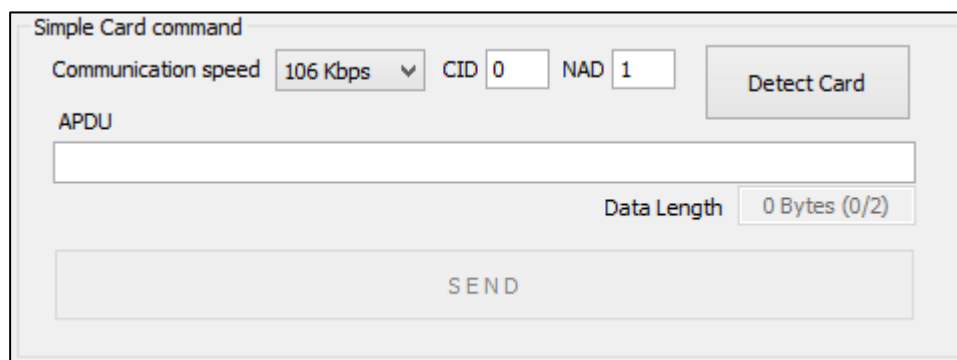
Analysis for [Detect Card]



<Figure 3-9>

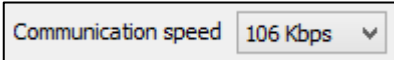
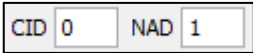
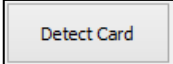
Analysis for [RATS]

3.5 Simple Card Command



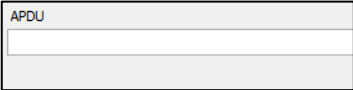
<Figure 3-10>

Simple Card Command

- : Select the communication speed, 106/212/424/848 kbps.
- : Set the CID and NAD values for card communication.
- : Click to execute the "Detect Card" command.

DualCardDII API : DE_FindCard([PORT],[BAUD],[CID],[NAD],[OPTION],[RLEN],[RBUF]);

DUALi RW Protocol : 4C[BAUD][CID][NAD][OPTION]

- : Input the APDU. The PCB and CID are automatically set with the default value (0) and added into the user ulcommand (APDU).

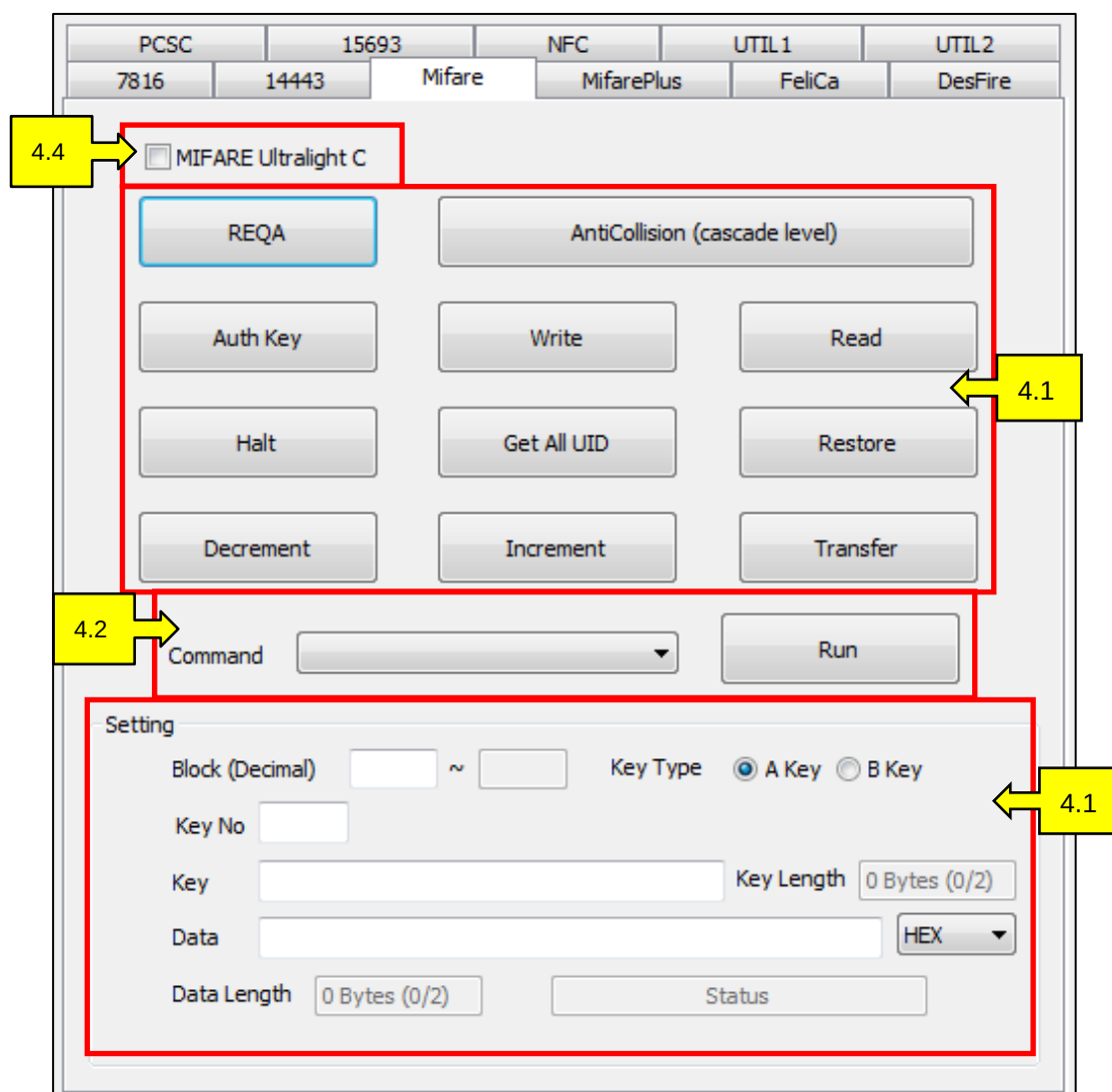
- : Click to transmit the APDU

DualCardDll API : DE_APDU([PORT],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol : 61[DATA]

4. Mifare

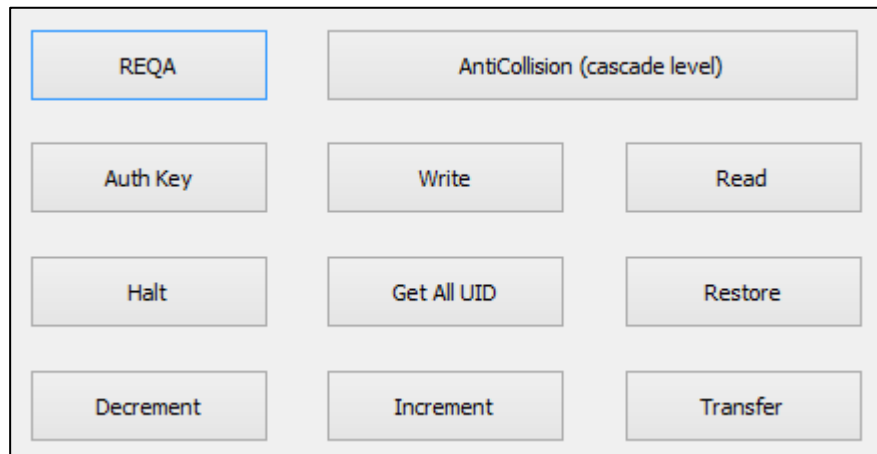
The Mifare tab, <Figure 4-1>, supports communication with NXP's contactless memory card, Mifare. This tab allows users to test all functions in the Mifare specification. Mifare cards (up to layer 2) use the same protocol as ISO 14443 Type A and use their own protocol from layer 3. Mifare specifications require an authentication process to access the memory of Mifare cards. By providing a 6 byte key the system gives permission to read or write data inside the memory. Each block is made of 16 bytes and the data process has to follow the same 16 byte format. Refer to NXP's specification for detailed technical information.



<Figure 4-1>

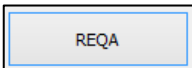
Mifare Tab

4.1 Button Type Command



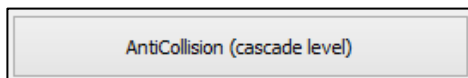
<Figure 4-2>

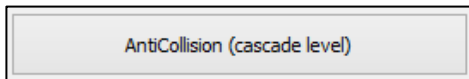
Button Type Command

- : Click to execute a “Request” command. A response will be displayed in the log, <Figure 1-7>.

DualCardDII API : `DEA_Idle_Req([PORT],[RLEN],[RBUF]);`

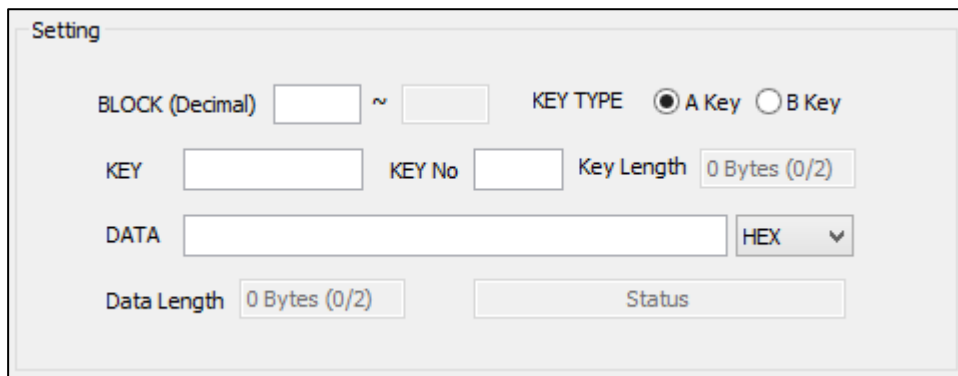
DUALi RW Protocol : 21



- : Click to execute an “AntiCollision” or “Select” command. A response will be displayed in the log, <Figure 1-7>.

DualCardDII API : `DEA_AntiSelLevel([PORT],[RLEN],[RBUF]);`

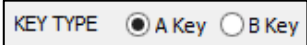
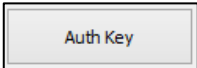
DUALi RW Protocol : 3D



<Figure 4-3>

Mifare Setting

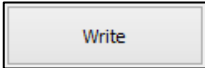
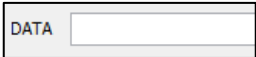
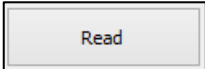
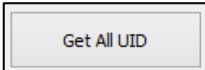
Steps for the Authentication process. (See Example 14.2)

1. : Input the block number
2. : Select a key type for authentication
3. : Input the key value to access a selected block (Hexadecimal, 6byte)
- : Click to execute an authorization command. A response will be displayed in the log, <Figure 1-7>.

DualCardDII API : DEA_Authkey([PORT],[MODE],[KEY],[BLOCK NO]);

DUALi RW Protocol : 30[MODE][KEY][BLOCK NO]

The following commands must be executed after the Authentication process.

- : Click to execute a "Write" command.
- : Input data on <Figure 4-3> and click "Write"
- DualCardDII API : DEA_Write([PORT],[BLOCK NO],[SLEN],[SBUF]);
- DUALi RW Protocol : 28[BLOCK NO][DATA]
- : Click to execute a "Read" command
- : Input the block number, in <Figure 4-3>, and click "Read" to get a response.
- : Click to read a block number. A 16 bytes response will be displayed if the action was successful.
- DualCardDII API : DEA_Read([PORT],[BLOCK NO],[RLEN],[RBUF]);
- DUALi RW Protocol : 27[BLOCK NO]
- : Click to execute a "Halt" command to stop responses. The Mifare card will not respond until it receives a "Wakeup" request.
- DualCardDII API : DEA_Halt([PORT]);
- DUALi RW Protocol : 26
- : Click to read the UID of a card in the RF field.
- DualCardDII API : DEA_Req_Select([PORT],[MODE],[RLEN],[RBUF]);
- DUALi RW Protocol : 390

- Restore

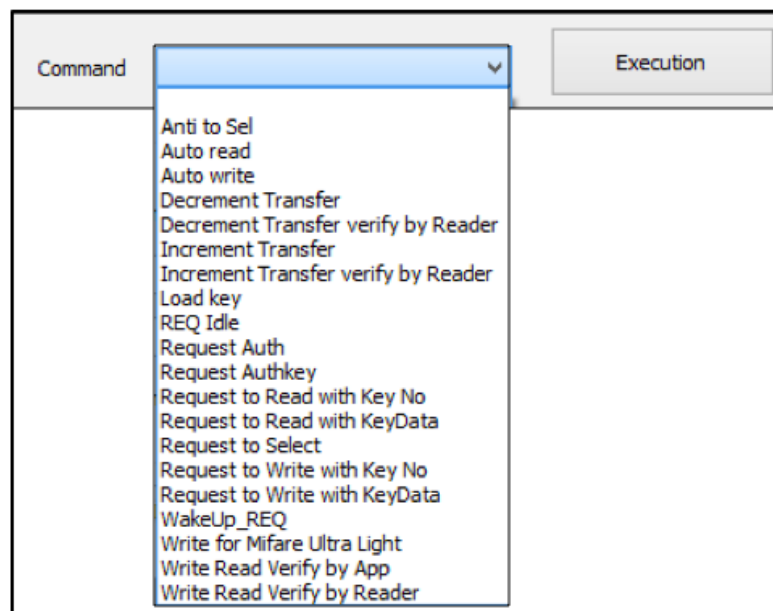
 : Click to execute a “Restore” command for the Value block.
 DualCardDII API : `DEA_Restore([PORT],[BLOCK NO]);`
 DUALi RW Protocol : `2D[BLOCK NO]`
- Decrement

 : Click to execute a “Decrement” command for the Value block. (See Example 14.2)
 DualCardDII API : `DEA_Decrement([PORT],[BLOCK NO],[VALUE]);`
 DUALi RW Protocol : `2A[BLOCK NO][VALUE]`
- Increment

 : Click to execute an “Increment” command for the Value block. (See Example 14.2)
 DualCardDII API : `DEA_Increment ([PORT],[BLOCK NO],[VALUE]);`
 DUALi RW Protocol : `29[BLOCK NO][VALUE]`
- Transfer

 : Select to execute a Transfer command for the Value block.
 DualCardDII API : `DEA_Transfer([PORT],[BLOCK NO]);`
 DUALi RW Protocol : `2E[BLOCK NO]`

4.2 Combo Box Type Command



<Figure 4-4>

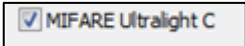
Combo Box Type Command

From the Mifare command list, select a command to execute and click [Execution]. (See Annex 13.1)

4.3 MIFARE Ultralight C



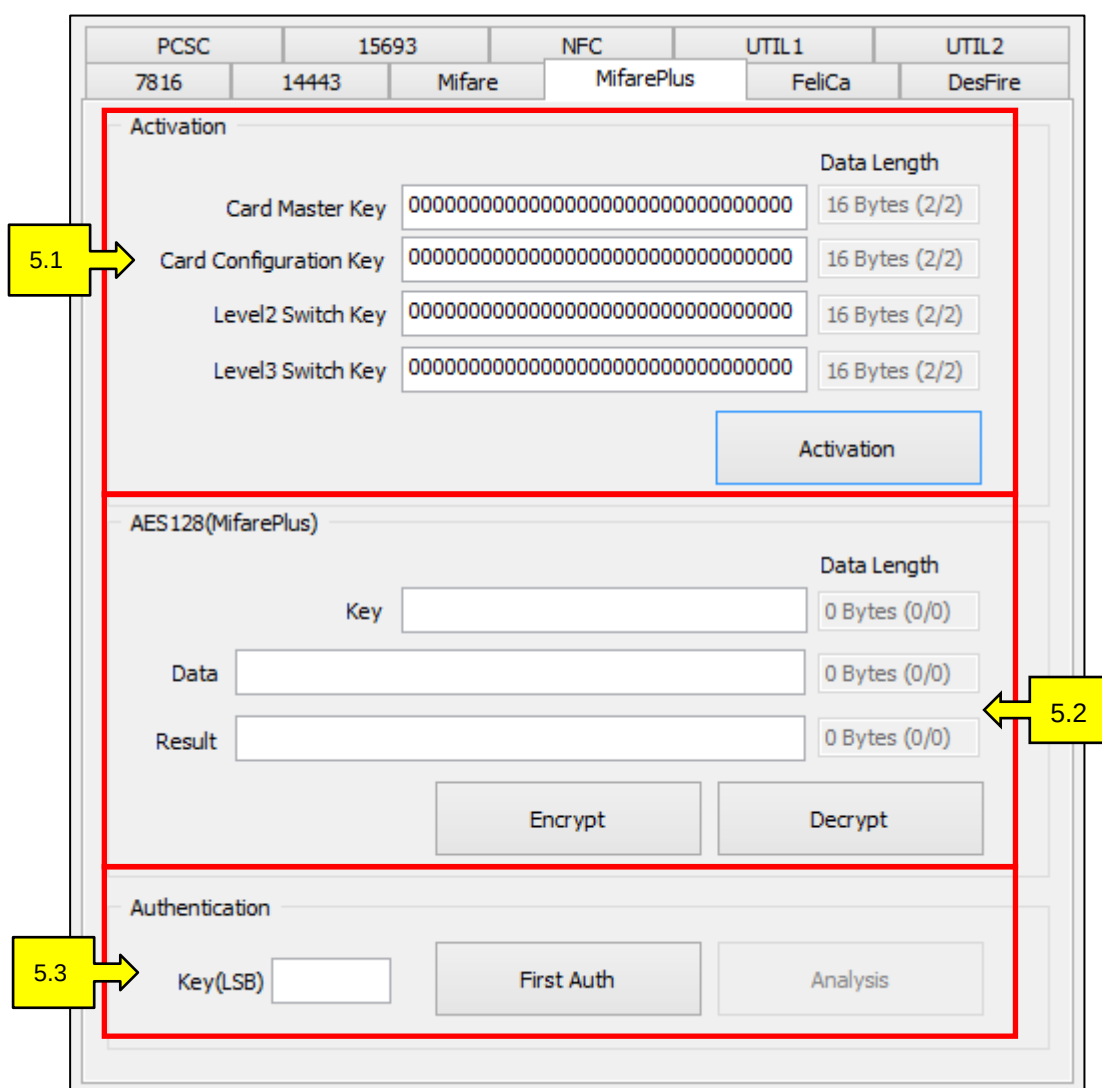
<Figure 4-5>

- : Click to active MIFARE Ultralight C block. It only activates for [REQA], [AntiCollision], [Auth Key], [Write], [Read], and [Halt] commands.
- After the [Auth Key] is selected, the user can use the Key function.

5. MifarePlus

The MifarePlus tab, <Figure 5-1>, supports communication with NXP's MifarePlus. MifarePlus has a more secure function in comparison to Mifare. A unique feature for MifarePlus is that the system will not respond until personalization is conducted.

- MifarePlus level 1 cards are compatible with the Mifare cards.
- MifarePlus level 2 cards use the AES128 algorithm for the authentication process.
- MifarePlus level 3 cards have adopted the ISO 14443 Type A –APDU format.



PCSC	15693	NFC	UTIL 1	UTIL 2
7816	14443	Mifare	MifarePlus	FeliCa

Activation

Card Master Key	00000000000000000000000000000000	Data Length	16 Bytes (2/2)
Card Configuration Key	00000000000000000000000000000000	Data Length	16 Bytes (2/2)
Level2 Switch Key	00000000000000000000000000000000	Data Length	16 Bytes (2/2)
Level3 Switch Key	00000000000000000000000000000000	Data Length	16 Bytes (2/2)

Activation

AES128(MifarePlus)

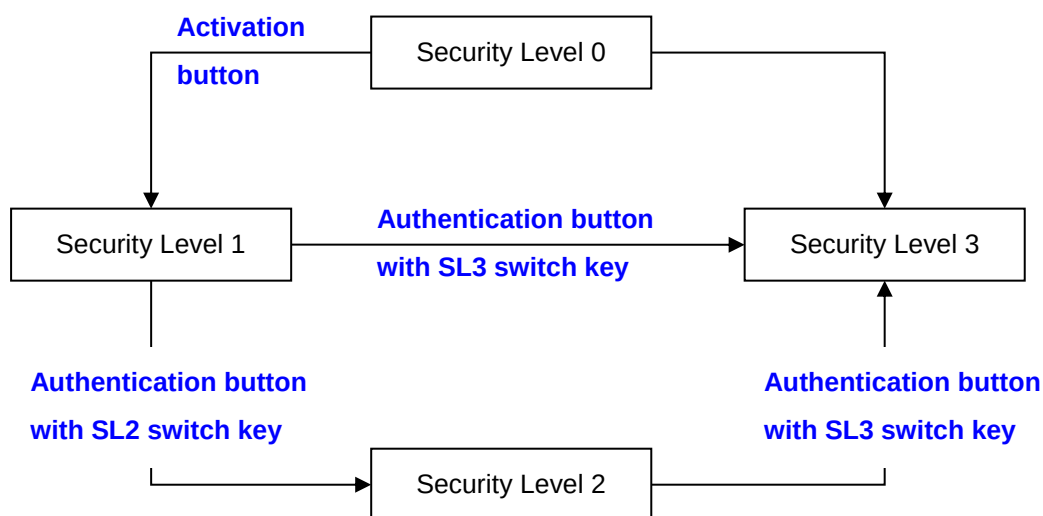
Key		Data Length	0 Bytes (0/0)
Data		Data Length	0 Bytes (0/0)
Result		Data Length	0 Bytes (0/0)

Encrypt Decrypt

Authentication

Key(LSB) First Auth Analysis

<Figure 5-1>
MifarePlus Tab

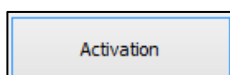


<Figure 5-2>
Security Level Switching

5.1 Activation

Activation		Data Length
Card Master Key	00000000000000000000000000000000	16 Bytes (2/2)
Card Configuration Key	00000000000000000000000000000000	16 Bytes (2/2)
Level2 Switch Key	00000000000000000000000000000000	16 Bytes (2/2)
Level3 Switch Key	00000000000000000000000000000000	16 Bytes (2/2)
Activation		

<Figure 5-3>
MifarePlus Activation



- Click to convert the Security level 0 (default) card to a Security level 1 card. The Security Level 1 card is compatible with the Mifare card.
- Fill out each key (default =0000000...000000) and click "Activation". It will process as: [REQA] -> [ANTICOLLISION] -> [SELECT] -> [RATS] -> [WRITE PERSO for each key] -> [COMMIT PERSO]
- The activation process sets: [Card Master key], [Card Configuration key], [Level 2 switch key], [Level 3 switch key]. It only sets keys which are mandatory to activate the card. Other keys

can be changed after the activation process and authentication.

- There are “x” & “s” versions of the MifarePlus cards. If using the “x” version, the [Level 2 switch key] is mandatory for activation. However, if using the “s” version, the [Level 2 switch key] is not necessary because the “s” version does not support Level 2.
- After the activation process, the Mifare Plus card becomes a Level 1 card (same as Mifare card). Users can use the “Mifare” tab for the next process (card read & write). Please refer to section 4, Mifare, in the User Manual

DualCardDII API :

REQA : DEA_Idle_Req([PORT],[RLEN],[RBUF]);

ANTICOLLISION, SELECT : DEA_AntiSelLevel([PORT],[RLEN],[RBUF]);

RATS, WRITE PERSO, COMMIT PERSO :

DEA_Transparent([PORT],[SLEN],[SBUF],[TIMEOUT],[RLEN],[RBUF]);

DUALi RW Protocol :

REQA : 21

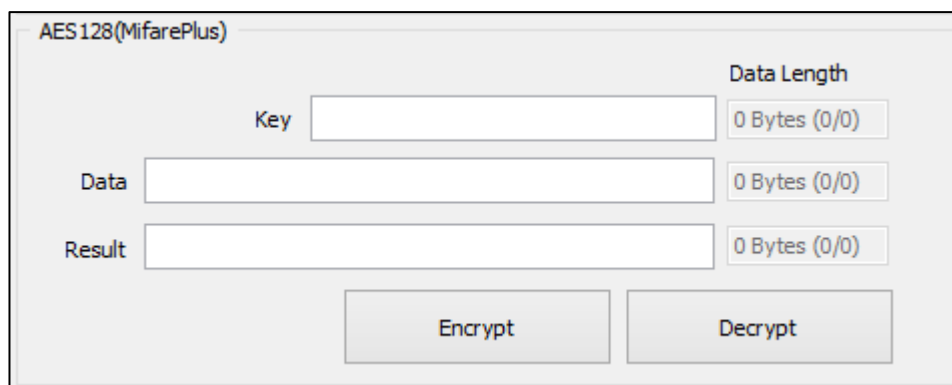
ANTICOLLISION, SELECT : 3D

RATS : 41E080[TIMEOUT]

WRITE PERSO : 41[PCB]A8[KEY NO][KEY][TIMEOUT]

COMMIT PERSO :41[PCB]AA[TIMEOUT]

5.2 AES128

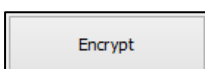


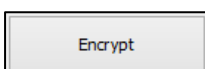
The screenshot shows a software interface titled "AES128(MifarePlus)". It contains three input fields: "Key", "Data", and "Result". To the right of each field is a "Data Length" indicator, all of which show "0 Bytes (0/0)". Below the input fields are two buttons: "Encrypt" and "Decrypt".

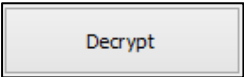
<Figure 5-4>

AES128

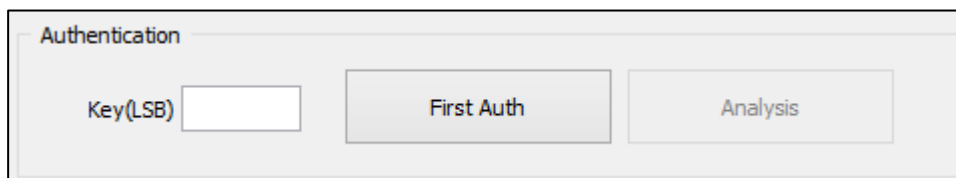
- MifarePlus cards use the AES128 algorithm for authentication.
- The AES 128 algorithm is necessary for authentication and communication with Level 2 and Level 3 cards.



- : Click after inputting [Key] and [Data]. Results will be displayed in the [Result] field after the data encryption uses the AES128 Algorithm.

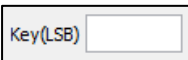
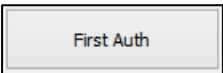
- : Click after inputting [Key] and [Data]. Results will be displayed in the [Result] box after the data decryption uses the AES128 Algorithm.

5.3 Authentication



<Figure 5-5>

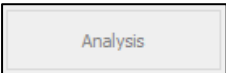
MifarePlus Authentication

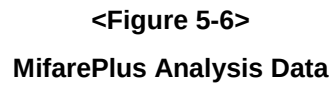
- : Input the Key to authenticate.
 - : Input the Key, in <Figure 5-4>, for selected Key(LSB).
 - : Click to process the authentication.
 - Example: Place a Level 3 card on the DE-620, input the Card master key into [Key] box, and input the key for selected key [0090]. Select [First Auth] to process authentication for the Card master key.
- DualCardDII API :**
- ```

REQA : DEA_Idle_Req([PORT],[RLEN],[RBUF]);
ANTICOLLISION, SELECT : DEA_AntiSelLevel([PORT],[RLEN],[RBUF]);
RATS, FIRST AUTH1, 2 :
DEA_Transparent([PORT],[SLEN],[SBUF],[TIMEOUT],[RLEN],[RBUF]);

```
- DUALi RW Protocol :**
- ```

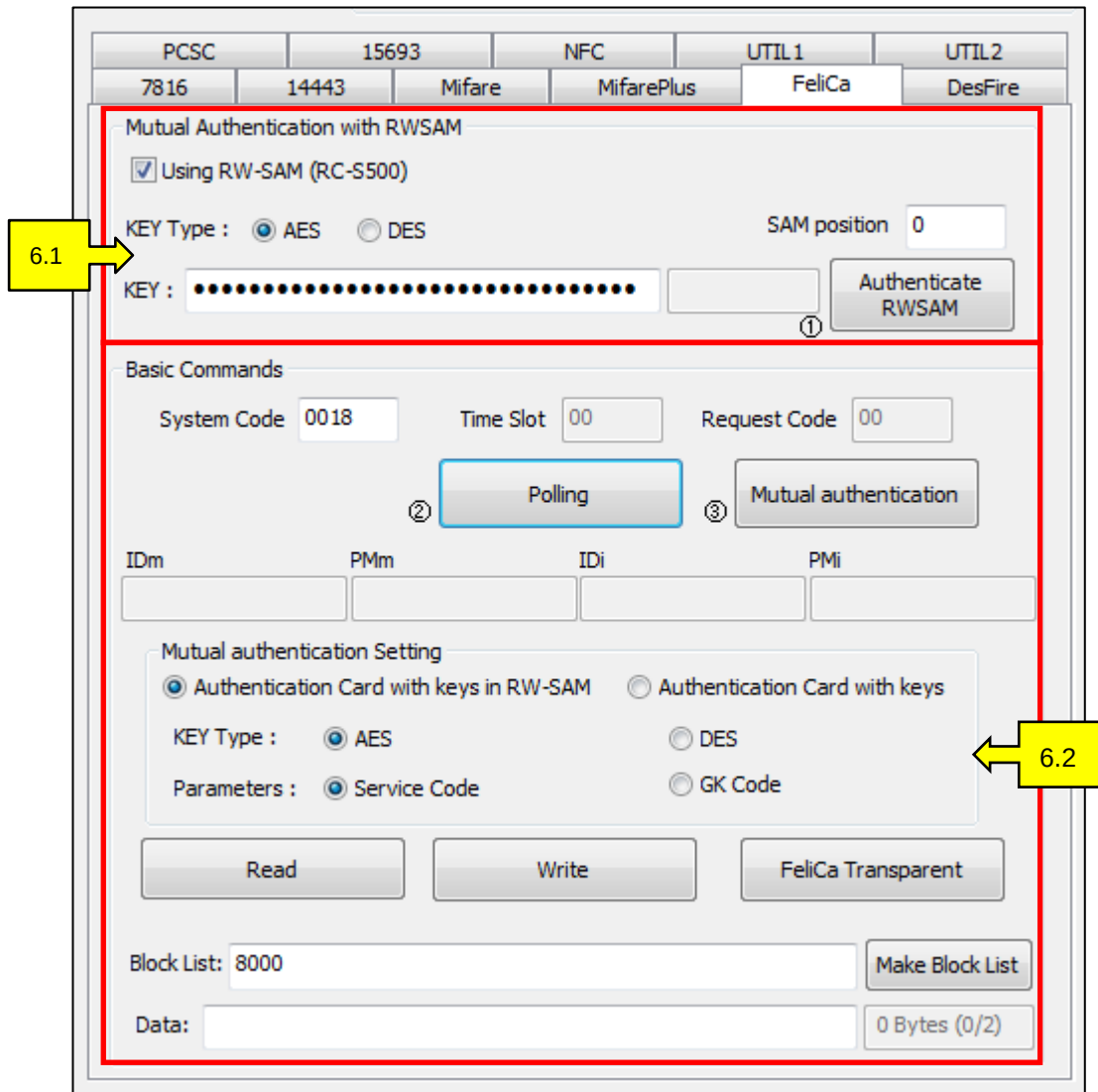
REQA : 21
ANTICOLLISION, SELECT : 3D
RATS : 41E080[TIMEOUT]
FIRST AUTH1 : 41[PCB]70[KEY NO]06000000000000[TIMEOUT]
FIRST AUTH2 : 41[PCB]72[CIPHER DATA][TIMEOUT]

```
- : Click to see the Analysis Data, <Figure 5-6>. This button will be activated when Authentication is successful.



6. FeliCa

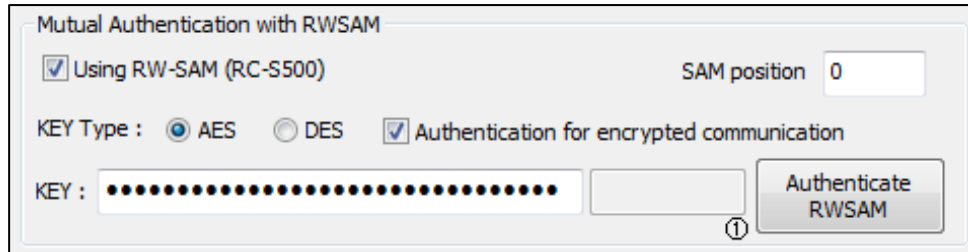
The FeliCa tab, <Figure 6-1>, supports communication with Sony's FeliCa cards. FeliCa cards use the sim types, RW-SAM and RC-S500, which allow higher security during transactions. This program is based on SONY's SAMPLE 5 cards for secured logic and keys. Refer to Sony's FeliCa specification for detailed technology information.



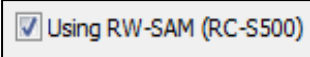
<Figure 6-1>

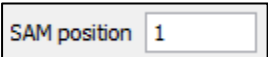
FeliCa Tab

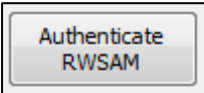
6.1 Mutual Authentication with RW-SAM



**<Figure 6-2>
RWSAM**

- : Check the box to use encrypted communication with RC-S500 SAM. (For encrypted communication use the RW-SAM).

- : Input the SAM slot position.

- : Click to execute a mutual authentication. A response will be displayed.
- (This is a batch function for mutual authentications)

DualCardDII API :

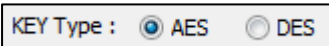
DE_FeliCa_RWSAMMutualAuthInDES ([PORT], [SLOTNUM] ,[ENCMODE], [KEY], [RLEN], [RBUF])

DE_FeliCa_RWSAMMutualAuthInAES ([PORT], [SLOTNUM] ,[ENCMODE], [KEY], [RLEN], [RBUF])

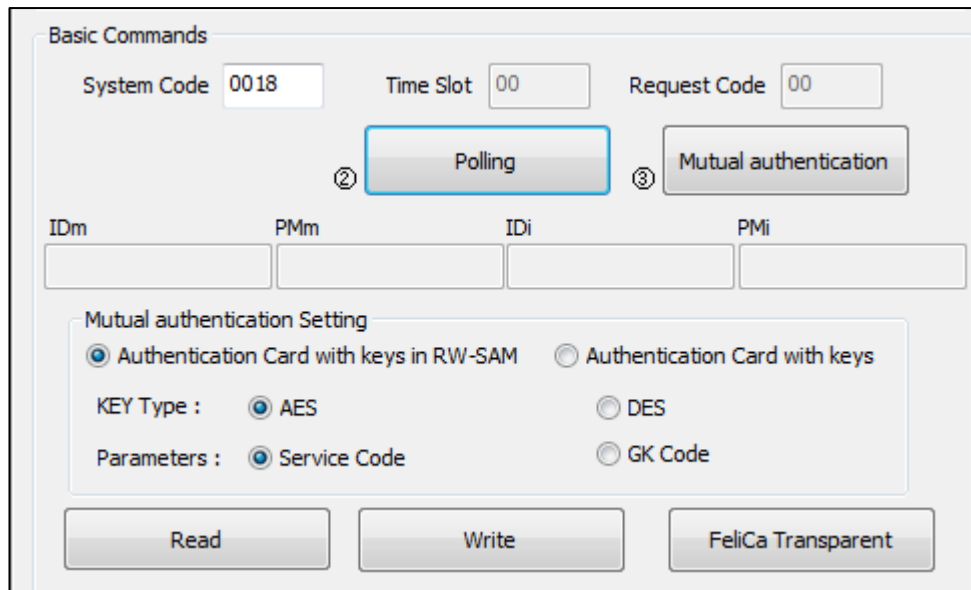
DUALi RW Protocol :

5A00[SLOT NO] ,[ENCMODE], [KEY]

5A01[SLOT NO] ,[ENCMODE], [KEY]

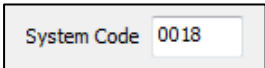
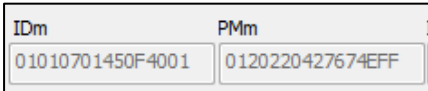
-  : Select the authentication Key Type.

6.2 FeliCa Communication using RW-SAM



<Figure 6-3>

Communication Using RW-SAM

- : Input the system code before sending "Polling" commands.
- : Click to send "Polling" commands, which come from RW-SAM. A response will be displayed for IDm and PMm. The time slot default value is 00(zero).
- : A response from the card will be displayed as [IDm] and

[PMm] after a "Polling" command is executed.

DualCardDll API :

DE_FeliCa_Polling ([PORT], [SYSTEMCODE], [RLEN], [RBUF])

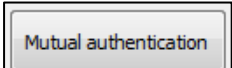
DE_FeliCa_PollingWithoutRWSAM ([PORT], [SYSTEMCODE], [TIMESLOT],

[REQUESTCODE], [RLEN], [RBUF])

DUALi RW Protocol :

5A10[SYSTEMCODE]

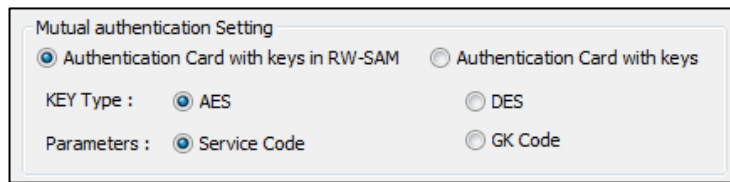
5B00[SYSTEMCODE] [TIMESLOT] ,[REQUESTCODE]

- : Click for a Mutual authentication between the RW-SAM and FeliCa card.

Six Modes of Mutual Authentication

The modes depend on the authenticated key type and path for the saving key.

1. Use AES key. Be aware of the system code and key version, select the options below.

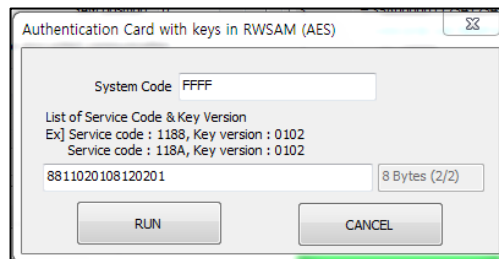


Mutual authentication Setting

☒ Authentication Card with keys in RW-SAM ☐ Authentication Card with keys

KEY Type : ☒ AES ☐ DES

Parameters : ☒ Service Code ☐ GK Code



Authentication Card with keys in RWSAM (AES)

System Code : FFFF

List of Service Code & Key Version
Ex] Service code : 1188, Key version : 0102
Service code : 118A, Key version : 0102

8811020108120201 8 Bytes (2/2)

RUN CANCEL

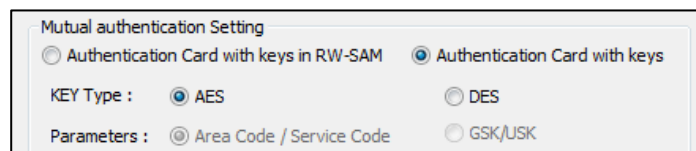
DualCardDII API :

DE_FeliCa_MutualAuthRWSAMInAES_SC ([PORT], [SYSTEMCODE], [nSERVICE], [SERVICECODE&VERSION LIST], [RLEN], [RBUF])

DUALi RW Protocol :

5A06[SYSTEMCODE], [nSERVICE], [SERVICECODE&VERSION LIST]

2. Use AES key. Be aware of the service code and Group key. Select the options below.

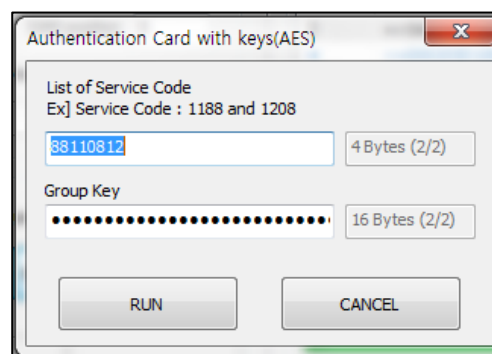


Mutual authentication Setting

☐ Authentication Card with keys in RW-SAM ☒ Authentication Card with keys

KEY Type : ☒ AES ☐ DES

Parameters : ☒ Area Code / Service Code ☐ GSK/USK



Authentication Card with keys(AES)

List of Service Code
Ex] Service Code : 1188 and 1208

38110812 4 Bytes (2/2)

Group Key
..... 16 Bytes (2/2)

RUN CANCEL

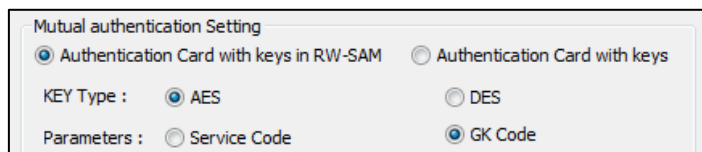
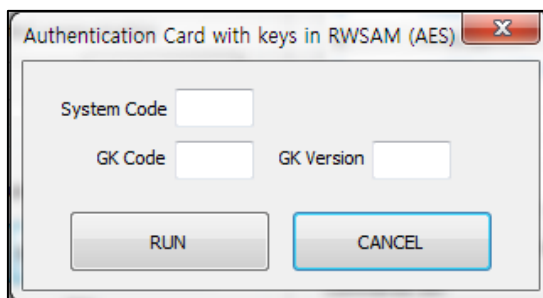
DualCardDII API :

DE_FeliCa_CardMutualAuthInAES_GK ([PORT], [nSERVICE], [SERVICECODE LIST],[GROUP KEY], [RLEN], [RBUF])

DUALi RW Protocol :

5A05 [nSERVICE], [SERVICECODE LIST],[GROUP KEY]

3. Use AES key. Be aware of system code , Group key code, and version, select the options below.

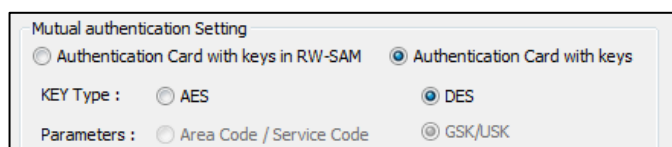
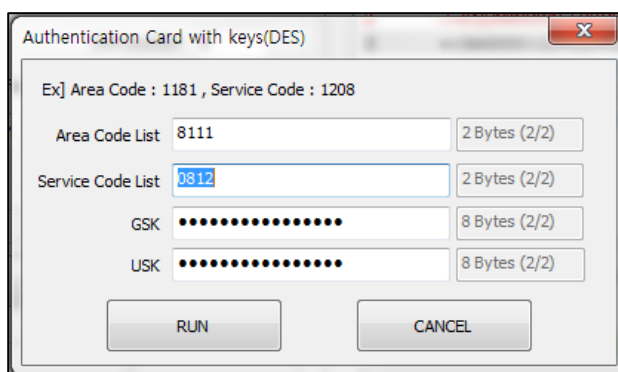
DualCardDll API :

DE_FeliCa_MutualAuthRWSAMInAES_GKC ([PORT], [SYSTEMCODE], [GROUP KEY CODE], [GROUP KEY VERSION], [RLEN], [RBUF])

DUALi RW Protocol :

5A07[SYSTEMCODE], [GROUP KEY CODE], [GROUP KEY VERSION]

4. Use DES key. Be aware of Area code, Service code, Group service key, User service key. Select the options below.

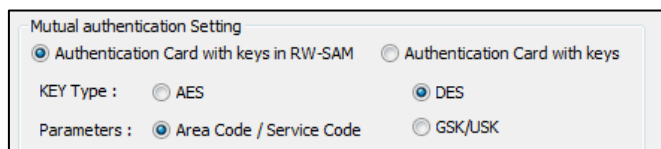
DualCardDll API :

DE_FeliCa_CardMutualAuthInDES_GSKUSK ([PORT], [nAREA], [AREACODE LIST], [nSERVICE], [SERVICECODE LIST],[GROUP SERVICE KEY], [USER SERVICE KEY], [RLEN], [RBUF])

DUALi RW Protocol :

5A02[nAREA], [AREACODE LIST], [nSERVICE], [SERVICECODE LIST],[GROUP SERVICE KEY], [USER SERVICE KEY]

5. Use DES key. Be aware of the Area code & version, Service code & version. Select the options below.

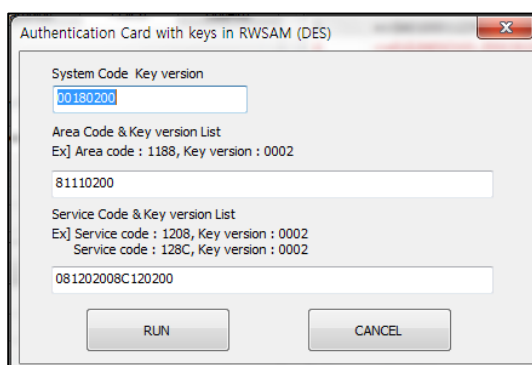


Mutual authentication Setting

☒ Authentication Card with keys in RW-SAM ☐ Authentication Card with keys

KEY Type : ☐ AES ☒ DES

Parameters : ☒ Area Code / Service Code ☐ GSK/USK



Authentication Card with keys in RWSAM (DES)

System Code Key version
00180200

Area Code & Key version List
Ex] Area code : 1188, Key version : 0002
81110200

Service Code & Key version List
Ex] Service code : 1208, Key version : 0002
Service code : 128C, Key version : 0002
081202008C120200

RUN CANCEL

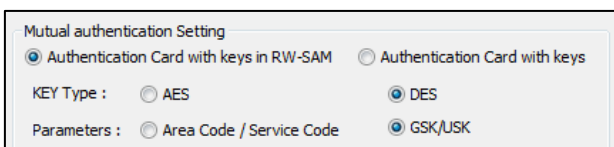
DualCardDll API :

DE_FeliCa_MutualAuthRWSAMInDES_ACSC ([PORT], [SYSTEMCODE], [SYSTEMKEY VERSION], [nAREA], [AREACODE&VERSION LIST], [nSERVICE], [SERVICECODE&VERSION LIST], [RLEN], [RBUF])

DUALi RW Protocol :

5A04[SYSTEMCODE], [SYSTEMKEY VERSION], [nAREA], [AREACODE&VERSION LIST], [nSERVICE], [SERVICECODE&VERSION LIST]

6. Use DES key. Be aware of Group service key & version, User service key & version. Select the options below.

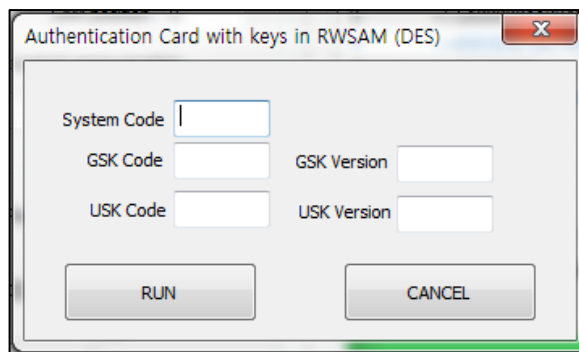


Mutual authentication Setting

☒ Authentication Card with keys in RW-SAM ☐ Authentication Card with keys

KEY Type : ☐ AES ☒ DES

Parameters : ☐ Area Code / Service Code ☒ GSK/USK

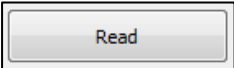
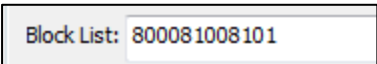


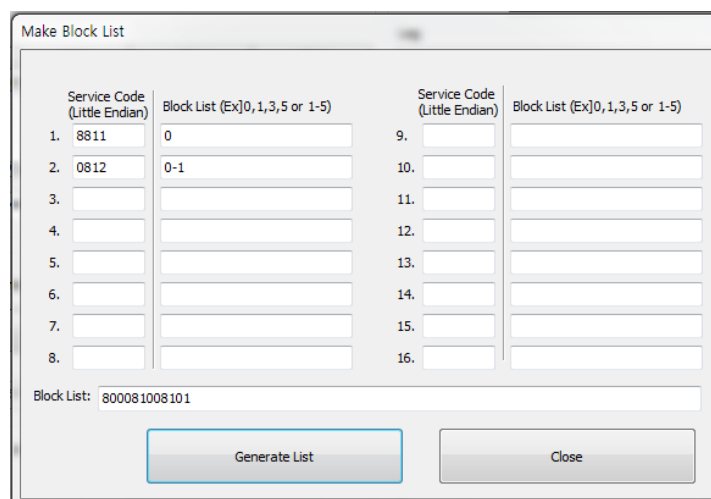
DualCardDII API :

DE_FeliCa_MutualAuthRWSAMInDES_GSKUSKVersion ([PORT], [SYSTEMCODE], [GSKCODE], [GSKVERSION], [USKCODE],[USKVERSION],[RLEN], [RBUF])

DUALi RW Protocol :

5A03[SYSTEMCODE], [GSKCODE], [GSKVERSION], [USKCODE],[USKVERSION]

- : Click to get data from the input block list.
- : Input [Block list] directly.
- : Click to create a block list with information of service codes and block numbers. Service codes. should be in the same order as when you used the mutual authentication.



<Figure 6-4>

Block List

DualCardDII API :

DE_FeliCa_ReadBlock([PORT], [nBLOCK], [BLOCKLIST],[RLEN], [RBUF])

DE_FeliCa_ReadBlockWithoutEnc([PORT], [nSERVICE], [SERVICECODELIST], [nBLOCK],

[BLOCKLIST], [RLEN], [RBUF])

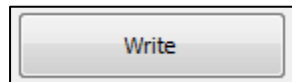
DE_FeliCa_ReadBlockWithoutEnc([PORT], [nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [RLEN], [RBUF])

DUALi RW Protocol :

5A11[nBLOCK], [BLOCKLIST]

5A13[nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST]

5B01[nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST]



- : Click to write data of input block list.

DualCardDII API :

DE_FeliCa_WriteBlock([PORT], [nBLOCK], [BLOCKLIST], [DATA], [RLEN], [RBUF])

DE_FeliCa_WriteBlockWithoutEnc([PORT], [nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [DATA], [RLEN], [RBUF])

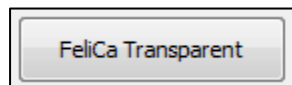
DE_FeliCa_WriteBlockWithoutEnc([PORT], [nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [DATA], [RLEN], [RBUF])

DUALi RW Protocol :

5A12[nBLOCK], [BLOCKLIST],[DATA]

5A14[nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [DATA]

5B02[nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [DATA]



- : Click to send commands to Felica cards.

DualCardDII API :

DEC_Transparent ([PORT],[DATA LEN],[DATA],[RLEN],[RBUF],[TIMEOUT]);

DUALi RW Protocol :

50[DATA LEN],[DATA] ,[TIMEOUT]

6.3 FeliCa Communication NOT using RW-SAM

Basic Commands

System Code

0018

Time Slot

00

Request Code

00

Polling

Mutual authentication

IDm

PMm

IDi

PMi

Mutual authentication Setting

☒ Authentication Card with keys in RW-SAM
 ☐ Authentication Card with keys

KEY Type :

☒ AES
 ☐ DES

Parameters :

☒ Service Code
 ☐ GK Code

Read

Write

FeliCa Transparent

Block List:

01CB11018000

Make Block List

Data:

0 Bytes (0/2)

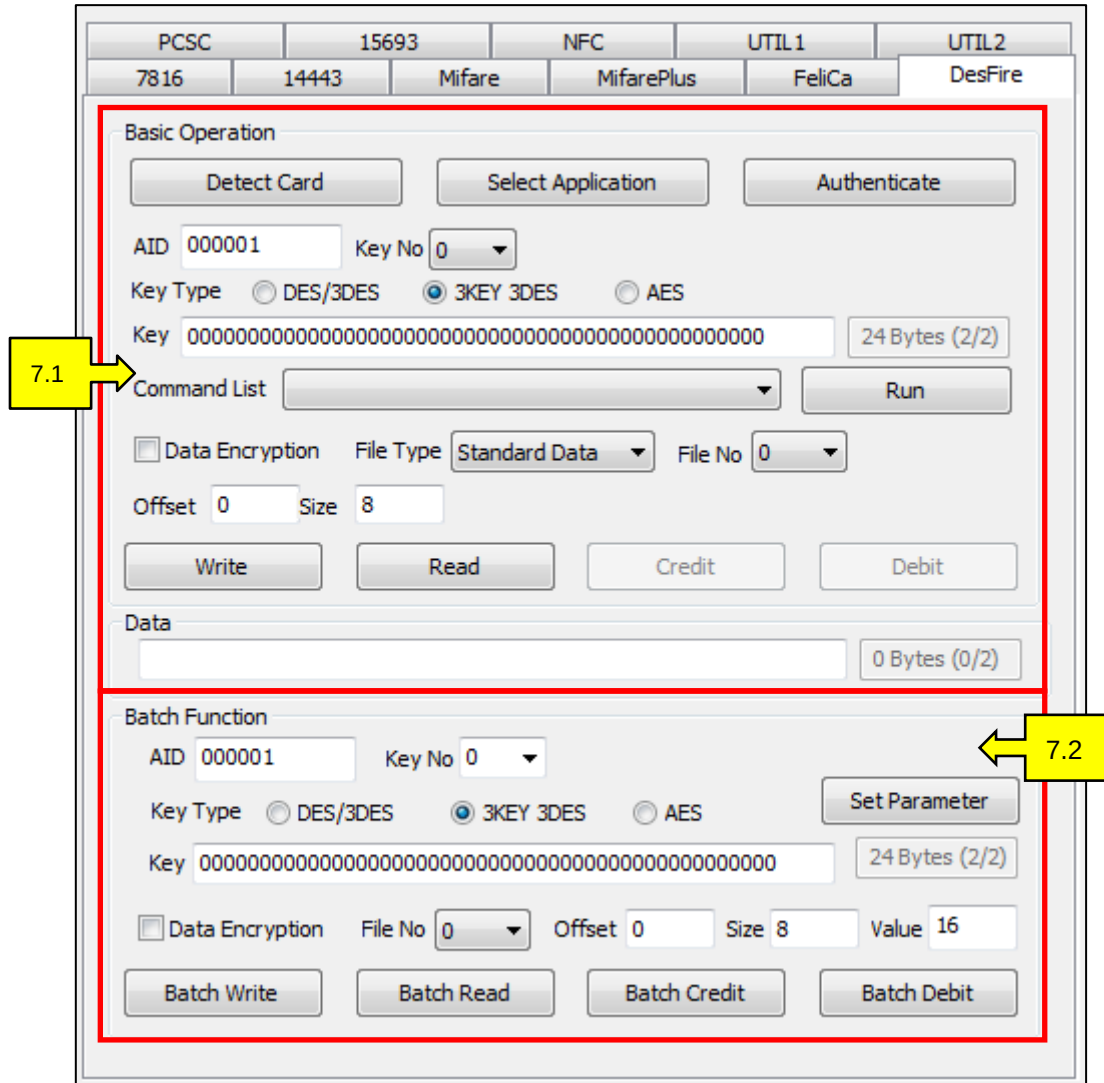
<Figure 6-5>

Communication Not Using RW-SAM

- After “Polling”, you can skip “Mutual Authentication” and proceed to the read/write function

7. DesFire

The DesFire tab, <Figure 7-1>, supports communication with NXP's DesFire cards.



<Figure 7-1>

DesFire Tab

7.1 Basic Operation

To use DesFire cards, first select the "AID" application then authenticate.

Basic Operation

Detect Card Select Application Authenticate

AID 000001 Key No 0

Key Type ☐ DES/3DES ☒ 3KEY 3DES ☐ AES

Key 00 24 Bytes (2/2)

Command List Run

☐ Data Encryption File Type Standard Data File No 0

Offset 0 Size 8

Write Read Credit Debit

Data

 0 Bytes (0/2)

<Figure 7-2>
DesFire Basic Operation

- : Click to confirm that the connected device is detecting cards

DualCardDll API :

```
DE_FindCard([PORT],[BAUD],[CID],[NID],[OPTION],[RLEN],[RBUF]);
```

DUALi RW Protocol :

4C00000000

-  : Select to choose an application ID

DualCardDll API :

```
DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);
```

DUALi RW Protocol :

(RW Protocol DesFire(150109) Refer to Chapter 5.1) 3A0200045A[AID]

- : Click after inputting the corresponding  key number, key

data , and AID (application ID) to authenticate. When the application ID is set to “000000”, the user will have full control of the PICC level.

DualCardDll API :

```
DE_DESFireAuthentication([PORT],[KEY NO],[KEY],[KEY LEN],[RBUF],[RLEN]);
```

```
DE_DESFireAuthentication AES([PORT],[KEY NO],[KEY],[KEY LEN],[RBUF],[RLEN]);
```

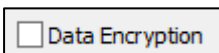
DUALi RW Protocol :

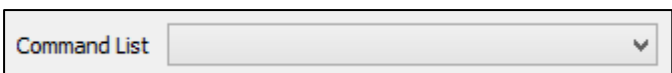
(Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

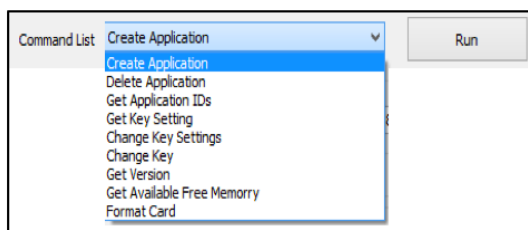
3A01[KEY NO][KEY]

3A05[KEY NO][KEY]

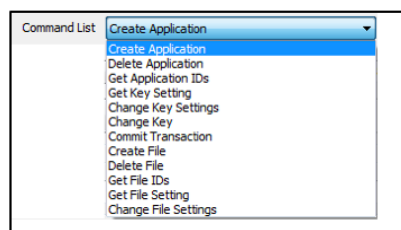
- : Select Key Type.

- : Check the box to encrypt commands (file level command)

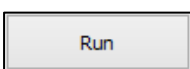
- : Select the appropriate command set. Different command lists will appear in the dropdown box when you select PICC level and the application level, as shown in <Figure 7.3> and <Figure 7.4>.

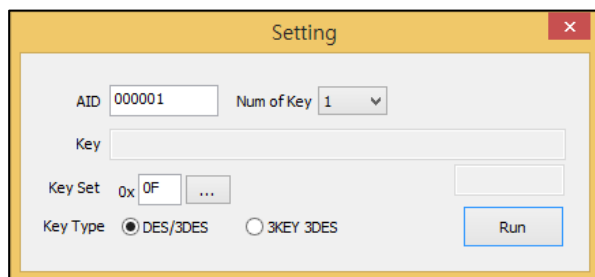


<Figure 7-3 Command for PICC Level>

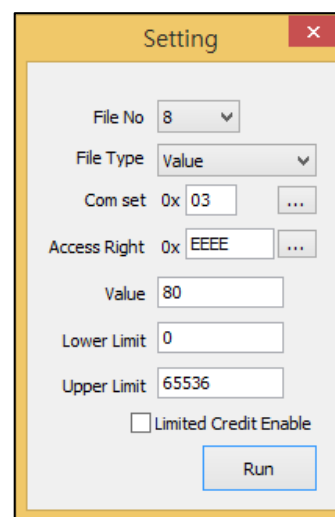


<Figure 7-4 Command for App Level>

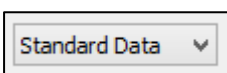
- : Click to execute the selected command in the command list. When additional information is required, the windows, <Figure 7-5> and <Figure 7-6>, will pop up.



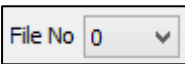
<Figure 7-5 Setting for Keys>

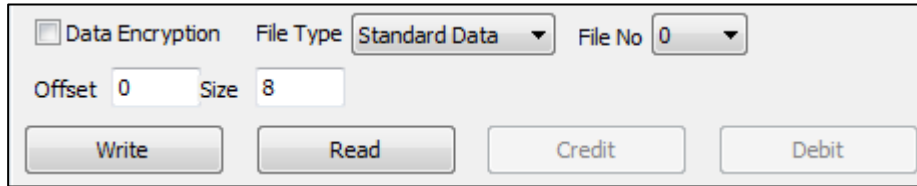


<Figure 7-6 Setting for Files>

- : Select a file type. Depending on the selected "File Type", different setting

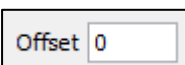
options will appear in the group box.

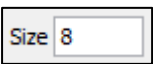
-  : Select the file number.

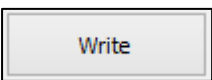


<Figure 7-7>

File Type, Standard Data

-  : Input the offset to read and write files.

-  : Input the size to read and write files.

-  : Click to write files.

DualCardDII API:

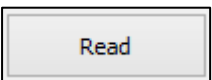
DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol:

3A02[FLAG]083D[FILE NO][OFFSET][FILE SIZE][DATA] data

3A02[FLAG]083B[FILE NO][OFFSET][FILE SIZE][DATA] record

3A02[FLAG]01AF

-  : Click to read files.

DualCardDII API :

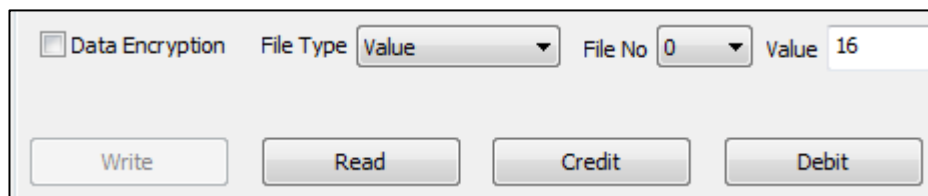
DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol :

3A02[FLAG]08BD[FILE NO][OFFSET][FILE SIZE] data

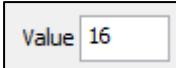
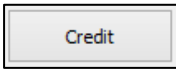
3A02[FLAG]08BB[FILE NO][OFFSET][FILE SIZE] record

3A02[FLAG]01AF



<Figure 7-8>

File Type, Value

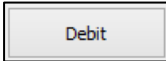
- : Input value of "Credit" or "Debit".
- : Click to add the number of credits, listed in the "Value" field.

DualCardDII API :

DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol :

3A02[FLAG]020C[FILE NO][VALUE]

- : Click to deduct the number of credits, listed in the "Value" field.

DualCardDII API :

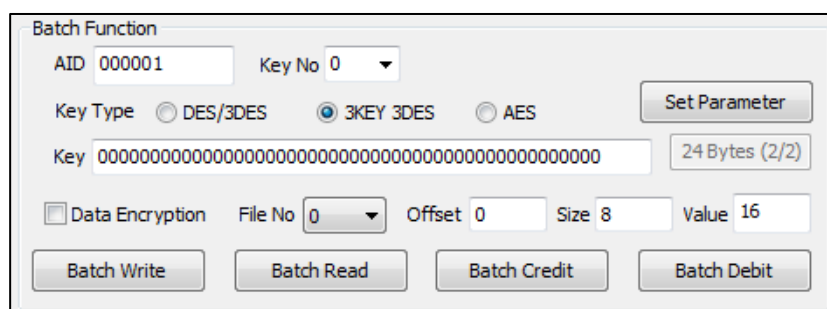
DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol :

3A02[FLAG]02DC[FILE NO][VALUE]

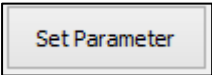
7.2 Batch Command

Using the "Batch" function is a faster and more convenient to execute multiple commands, especially when the user is not familiar with Desfire specifications and structure. Dualcard performs batch commands that shortcut the process from "Detect" to the "selected command protocol".



<Figure 7-9>

Batch Function

- : Click to set the Application ID, Key Number and Key value on the device.

DualCardDII API :

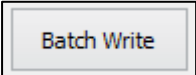
DE_DESFire_SetConfig_Batch([PORT], [AID], [KEY], [KEY LEN], [KEY NO]);

DE_DESFire_SetConfig_Batch_AES([PORT], [AID], [KEY], [KEY LEN], [KEY NO]);

DUALi RW Protocol : (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

3A03[AID][KEY][KEY NO]

3A06[AID][KEY][KEY NO]

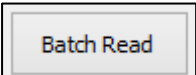
- : Click to execute the batch command from “Detect” file to “Write” file

DualCardDII API :

DE_DESFire_WriteFile_Batch ([PORT], [FLAG], [FILE NO], [OFFSET], [FILE SIZE], [DATA]);

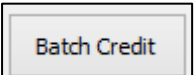
DUALi RW Protocol : (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

3A04[FALG]3D[FILE NO][OFFSET][FILE SIZE][DATA]

- : Click to execute the batch command from “Detect” file to “Read” file
- DualCardDII API : DE_DESFire_ReadFile_Batch ([PORT], [FLAG], [FILE NO], [OFFSET], [FILE SIZE], [DATA LEN], [DATA]);

DUALi RW Protocol : (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

3A04[FALG]BD[FILE NO][OFFSET][FILE SIZE]

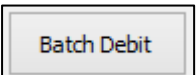
- : Click to execute the batch command from “Detect” file to “Credit”. This command adds the number of credits, listed in the “Value” field, to the value type. By using the “Batch Credit” option, you are conducting a single step instead of four steps (“Detect Card”, “Select Application”, “Authenticate”, and “Credit”).

DualCardDII API :

DE_DESFire_Credit_Batch ([PORT], [FLAG], [FILE NO], [VALUE]);

DUALi RW Protocol : (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

3A04[FALG]0C[FILE NO][OFFSET][FILE SIZE][DATA]

- : Click to execute the batch command from “Detect” file to “Debit”. This command subtracts the number of debits, listed in the “Value” field, to the value type. By using the “Batch Debit” option, you are conducting a single step instead of four steps (“Detect Card”, “Select Application”, “Authenticate”, and “Debit”).

DualCardDII API:

DE_DESFire_Debit_Batch ([PORT], [FLAG], [FILE NO], [VALUE]);

DUALi RW Protocol: (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

3A04[FALG]DC[FILE NO][OFFSET][FILE SIZE][DATA]

– Key Type ☐ DES/3DES ☒ 3KEY 3DES ☐ AES : Select Key Type

Example 14.4.2, Batch Command

☐ Use 3KEY3DES ☐ Data Encryption

Detect Card

AID : 000001

Select Application

Authenticate

Command List:
Commit Transaction

Key Setting : 0x 0F ...
Num of Keys : 01

Run

Analysis

File Type : Standard Data

File Number : 0

Offset : 0

File Size : 10

- [Batch Read]

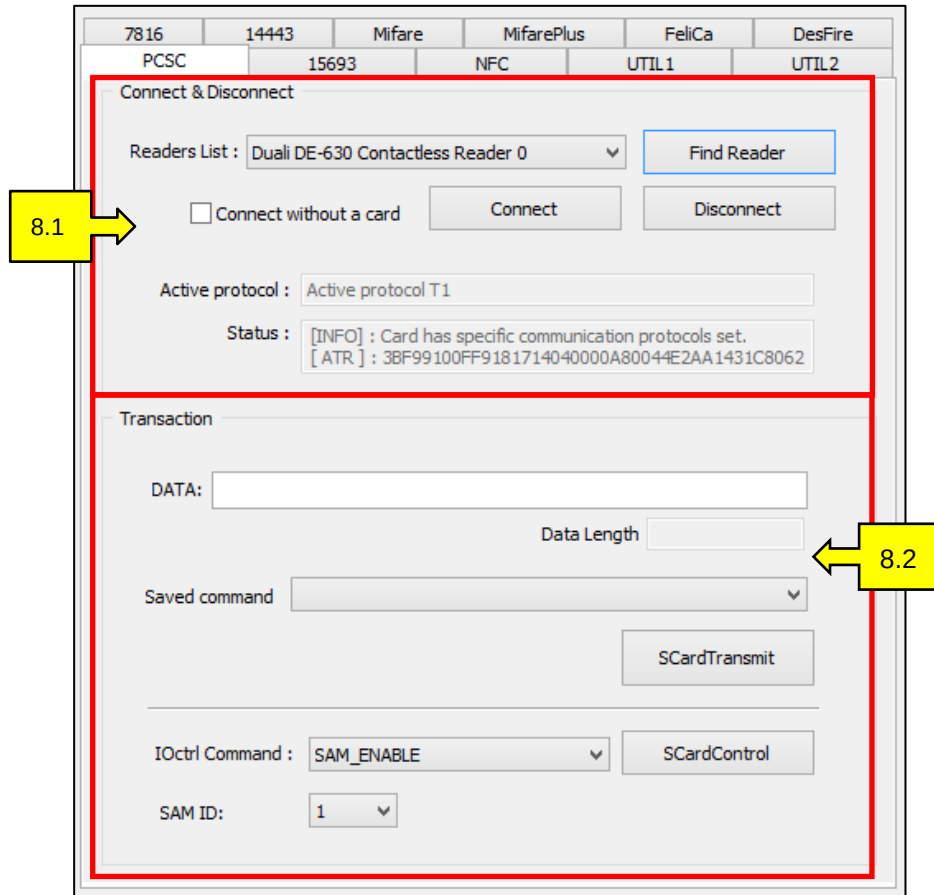
1. Set Application ID, Key Number and Key value.

2. Select "Set Parameter"

3. Select "Batch Read"

8. PCSC

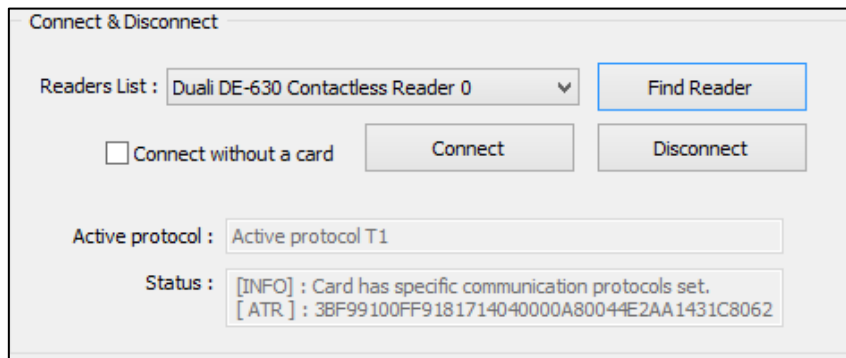
This tab, <Figure 8-1>, supports communication standards from Microsoft Window and PCSC.



<Figure 8-1>

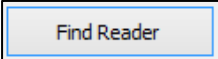
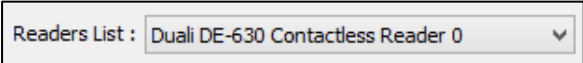
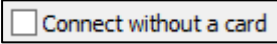
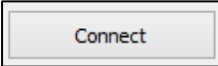
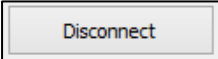
PCSC Tab

8.1 Connect and Disconnect

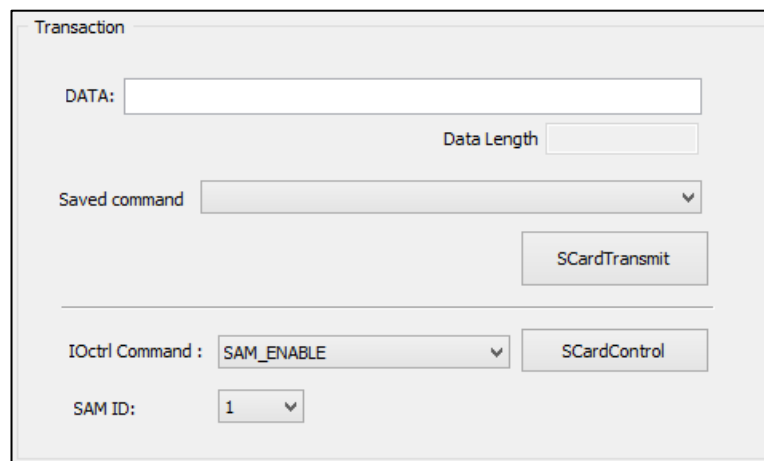


<Figure 8 -2>

PCSC Connect and Disconnect

- : Click to find the PCSC device that is connected to the PC.
- : Select a device that has been detected.
- : Check the box if an RF card is not being used during this process.
- : Click to connect the selected device. When a card is placed on the device, the “Status” will display the ATR value. If there is no RF card, select “Disconnect”.
- : Click to disconnect the PCSC device.

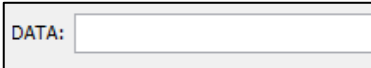
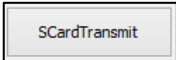
8.2 Transaction

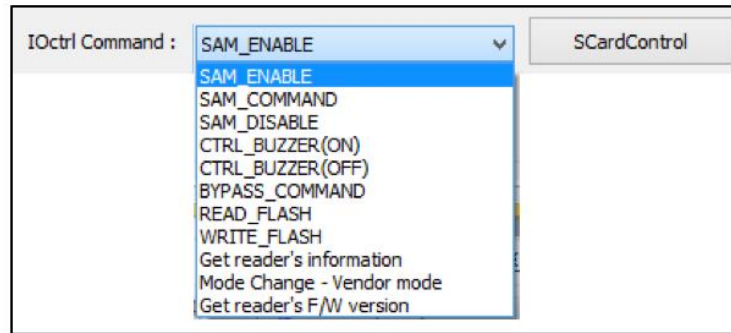


The Transaction dialog box contains the following fields and buttons:

- DATA:** A text input field for entering transaction data.
- Data Length:** A text input field for specifying the length of the data.
- Saved command:** A dropdown menu for selecting a saved command.
- SCardTransmit:** A button to execute the transaction.
- IOctrl Command:** A dropdown menu with the selected value **SAM_ENABLE**.
- SCardControl:** A button to execute the control command.
- SAM ID:** A dropdown menu with the selected value **1**.

<Figure 8-3>
PCSC Transaction

- : Input a command, this is optional.
- : Click to execute the data or command in the “Data” field.



<Figure 8-4>

IOctrl Command list

- To use the IOctrl Command from the PCSC communication specification:

SAM ID: ▼

: Select the SAM, only if there are more than 2 SAMs

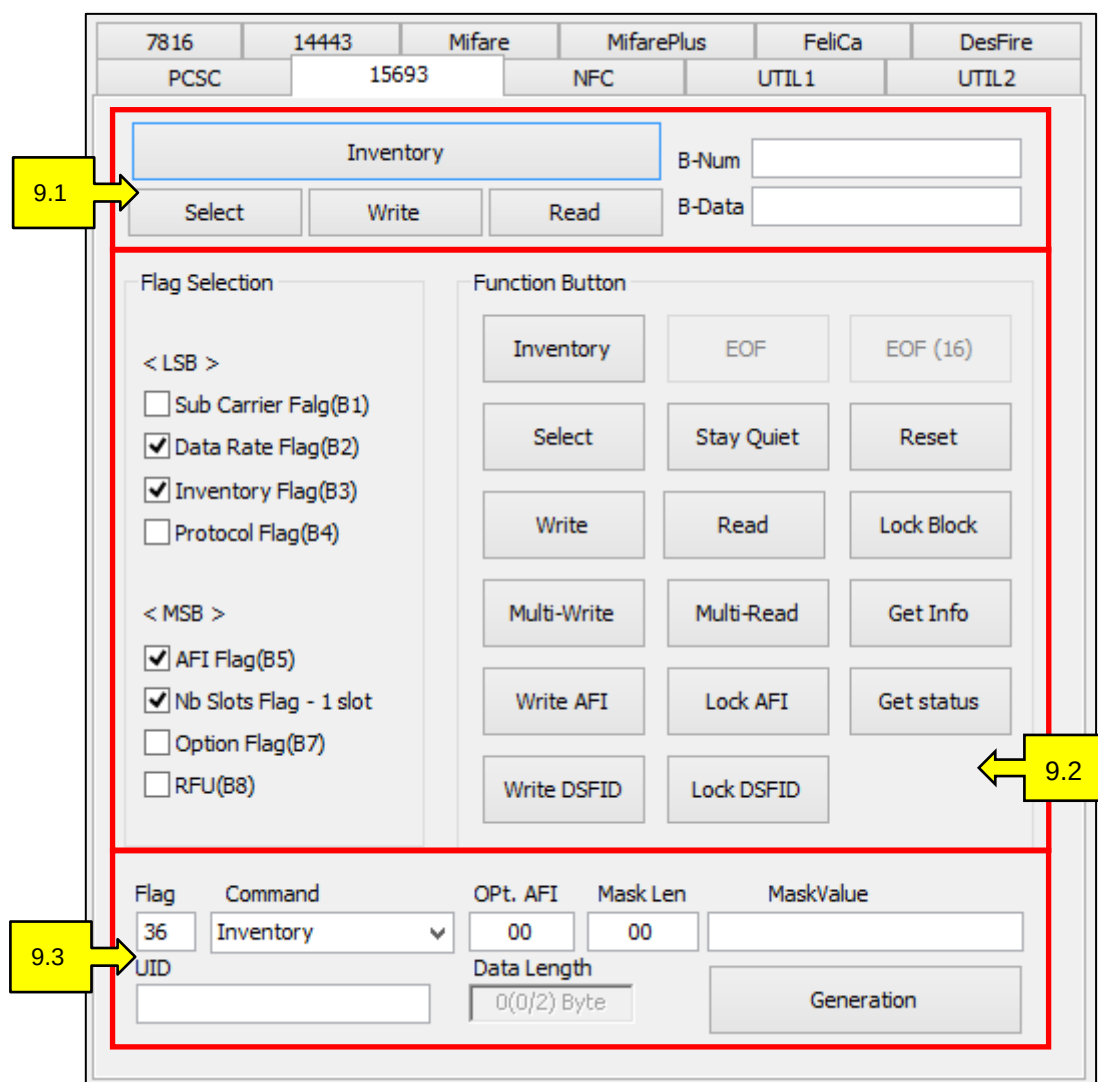
SCardControl

: Click to execute the IOctrl Command.

9. 15693 (Optional)

This tab, <Figure 9-1>, supports communication with ISO 15693. Refer to ISO 15693 or ISO 18000-3 specifications for detailed technical information.

*Contact the DUALi Lab to verify if your device supports the ISO 15693 protocol.



<Figure 9-1>

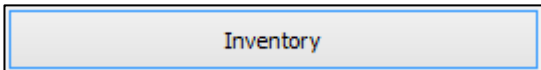
15693 Tab

9.1 Inventory & Select & Write & Read

Inventory			B-Num	
Select	Write	Read	B-Data	

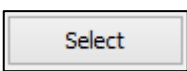
<Figure 9-2>

15693 Inventory

- : Click to execute the inventory command. This command reads multiple card UIDs at one time.

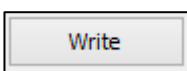
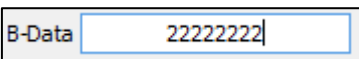
DualCardDII API : DED_Inventory([PORT],[FLAG],[RLEN],[RBUF]);

DUALi RW Protocol : 7001

- : Click to select the last detected card.

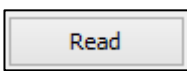
DualCardDII API : DED_Select([PORT],[RLEN],[RBUF]);

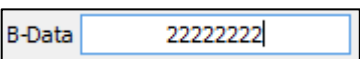
DUALi RW Protocol : 71

- : Click to write the B-Data, , in the B-Num, .

DualCardDII API : DED_Write([PORT],[BLOCK NO],[DATA],[RLEN],[RBUF]);

DUALi RW Protocol : 73[BLOCK NO][DATA]

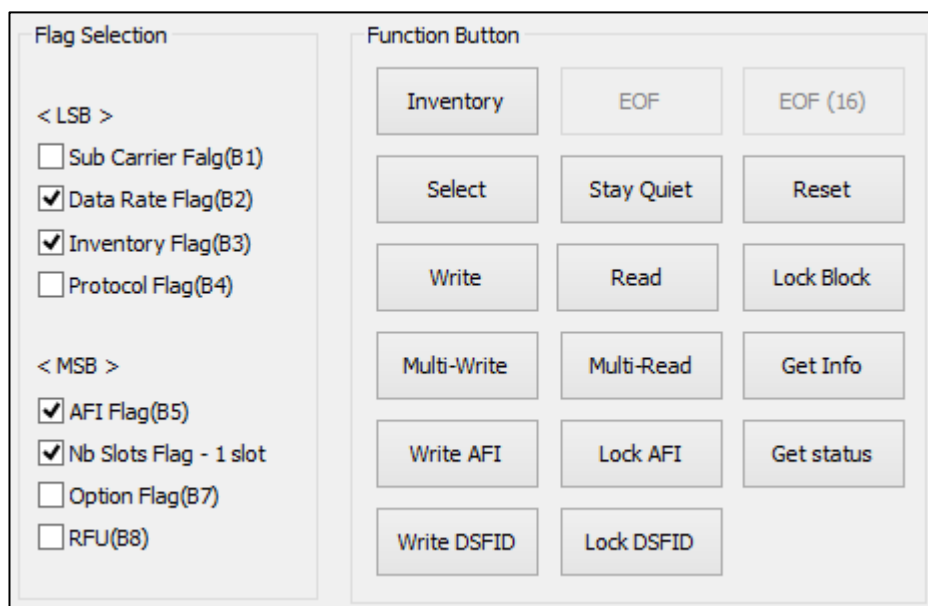
- : Click to read the B-Data in the B-Num, . The

B-Data will be displayed in .

DualCardDII API : DED_Read([PORT],[BLOCK NO],[RLEN],[RBUF]);

DUALi RW Protocol : 72[BLOCK NO]

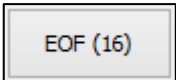
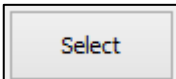
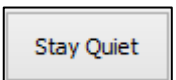
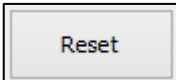
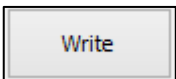
9.2 Select Flag & Create Command

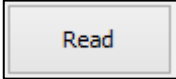
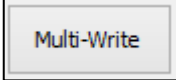
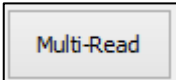
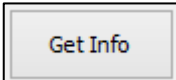
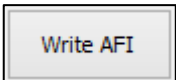
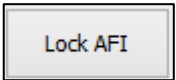
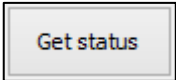
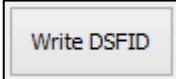


<Figure 9-3>

Flag Selection and Function Button

- The “Flag Selection” group box allows users to set flags for when devices send commands to card. Refer to <Figure 9-4>.
- The “he gure 9-4>.dis group box has a list ISO15693 commands. You can also select commands from a list, refer to <Figure 9-4>.

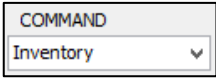
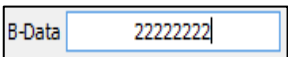
-  , DUALi RW Protocol : 74[FLAG]01[BLOCK NO][BLOCK COUNT]
-  , DualCardDII API : DED_Eof([PORT],[RLEN],[RBUF]);
DUALi RW Protocol : 78
-  , DualCardDII API : DED_Eof([PORT],[RLEN],[RBUF]);
DUALi RW Protocol : 78
-  , DUALi RW Protocol : 74[FLAG]25[UID]
-  , DUALi RW Protocol : 74[FLAG]02[UID]
-  , DUALi RW Protocol : 74[FLAG]26[UID]
-  , DUALi RW Protocol : 74[FLAG]21[UID][BLOCK NO][BLOCK COUNT]

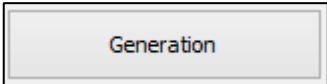
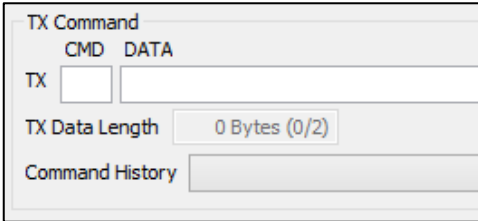
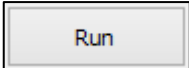
-  , DUALi RW Protocol : 74[FLAG]20[UID][BLOCK NO]
-  , DUALi RW Protocol : 74[FLAG]22[UID][BLOCK NO]
-  , DUALi RW Protocol : 74[FLAG]24[UID][BLOCK NO][BLOCK COUNT][BLOCK DATA]
-  , DUALi RW Protocol : 74[FLAG]23[UID][BLOCK NO][BLOCK COUNT]
-  , DUALi RW Protocol : 74[FLAG]2B[UID]
-  , DUALi RW Protocol : 74[FLAG]27[UID][BLOCK NO]
-  , DUALi RW Protocol : 74[FLAG]28[UID]
-  , DUALi RW Protocol : 74[FLAG]2C[UID][BLOCK NO][BLOCK COUNT]
-  , DUALi RW Protocol : 74[FLAG]29[UID][BLOCK NO]
-  , DUALi RW Protocol : 74[FLAG]2A[UID]

9.3 Command Execution

Flag	Command	Opt. AFI	Mask Len	MaskValue
36	Inventory	00	00	
UID		Data Length		
		0(0/2) Byte		Generation

<Figure 9-4>

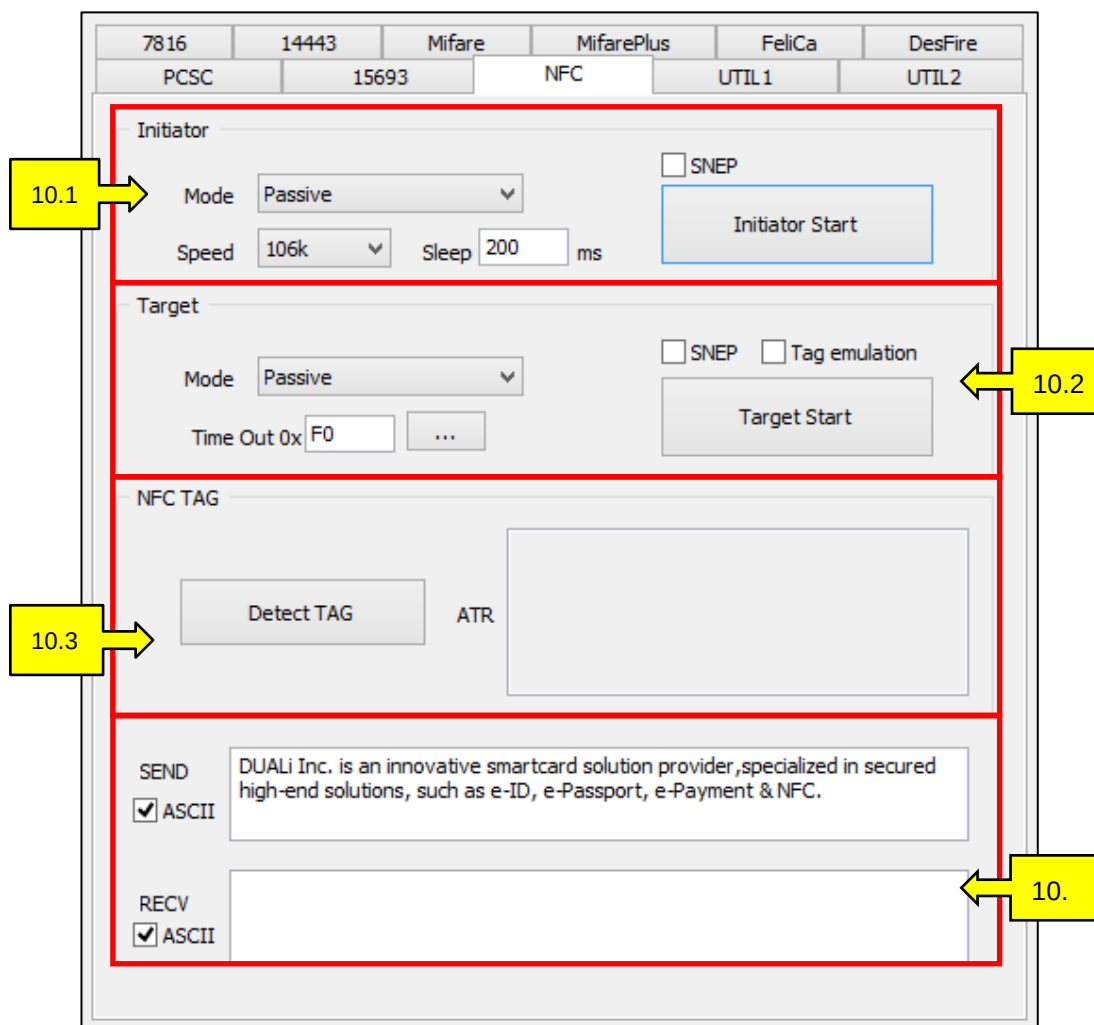
-  : Set flags for when the device sends commands to card
-  : Select commands, this optional. Otherwise, the “Function Button” group box has a list ISO15693 commands, refer to <Figure9-3>.
-  : The field where commands display responses.

- : Click after setting.
- : the command frame will be displayed in the log,
<Figure 1-7>.
- : Click to execute commands.

10. NFC

This tab, <Figure 10-1>, supports communication with ISO 18092 (Near Field Communication). ISO 18092 supports communication between Device to Device / Device to Card. Currently the communication speed supports 106, 212, and 424 kbps. For speeds of 106 kbps, the communication format is similar to that of ISO 14443 type A. For speeds of 212 and 424 kbps, the communication format is similar to that of FeliCa.

- Passive Communication: When one device creates an RF Field and the communication between devices are preset as either Initiator (creates RF field) or Target (receives power from Initiator).
- Active Communication: When both of the communication devices can create RF Fields and both devices can either be the Initiator or Target. The device which starts the communication is the Initiator and the other device is the Target.

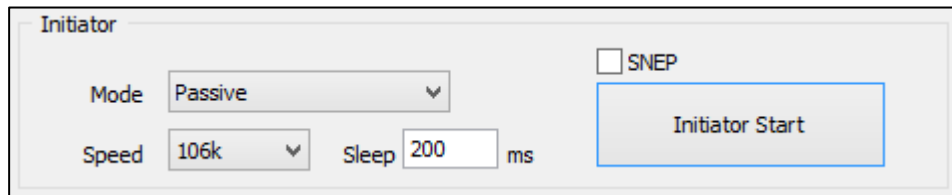


The screenshot shows the NFC configuration window with the following sections and callouts:

- 10.1** points to the **Initiator** section, which includes:
 - Mode: **Passive** (dropdown)
 - Speed: **106k** (dropdown)
 - Sleep: **200** ms
 - ☐ SNEP
 - Initiator Start** button
- 10.2** points to the **Target** section, which includes:
 - Mode: **Passive** (dropdown)
 - ☐ SNEP ☐ Tag emulation
 - Time Out 0x **F0** (input field)
 - Target Start** button
- 10.3** points to the **NFC TAG** section, which includes:
 - Detect TAG** button
 - ATR** (text area)
- 10.** points to the **SEND** and **RCV** sections, which include:
 - SEND** section with ☒ ASCII and a text area containing: "DUALi Inc. is an innovative smartcard solution provider, specialized in secured high-end solutions, such as e-ID, e-Passport, e-Payment & NFC."
 - RCV** section with ☒ ASCII and an empty text area.

<Figure 10-1>
NFC Tab

10.1 Initiator

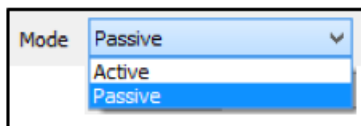


The image shows a configuration window titled "Initiator". It contains the following elements:

- Mode:** A dropdown menu currently set to "Passive".
- Speed:** A dropdown menu currently set to "106k".
- Sleep:** A text input field containing "200" followed by "ms".
- SNEP:** An unchecked checkbox labeled "SNEP".
- Initiator Start:** A button with the text "Initiator Start".

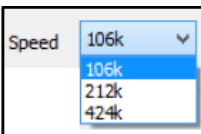
<Figure 10-2>
NFC Initiator

- An Initiator is the device that initiates the communication between NFC devices.



A dropdown menu for selecting the communication mode. The visible options are "Passive", "Active", and "Passive".

- : Select the communication mode of the Initiator.



A dropdown menu for selecting the communication speed. The visible options are "106k", "212k", and "424k".

- : Select the communication speed between the Initiator and Target.



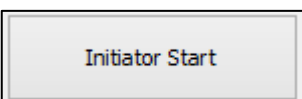
A text input field for setting the response waiting time, currently showing "200" followed by "ms".

- : Set the response waiting time.



An unchecked checkbox labeled "SNEP".

- : Check the box to send commands in SNEP format.



A button labeled "Initiator Start".

-

DualCardDll API :

```
LLC_Send_Text([PORT],0,[SLEN],[SBUF],[RLEN],[RBUF]);
```

```
DE_NFC_Init_INITIATOR([PORT],[MODE],[SPEED],[RLEN],[RBUF]);
```

```
DE_NFC_SendData([PORT],[NFC_INITIATOR_MODE],[SLEN],[SBUF],[RLEN],[RBUF]);
```

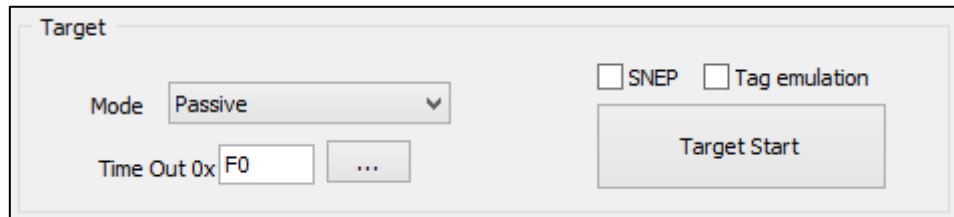
DUALi RW Protocol :

```
BC00[DATA]
```

```
94[MODE][SPEED]
```

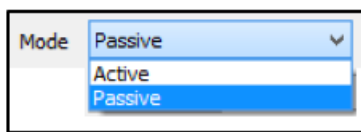
```
96[DATA]
```

10.2 Target

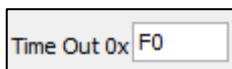


<Figure 10-3>
NFC Target

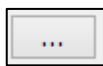
- A Target is the device that responds to the Initiator.



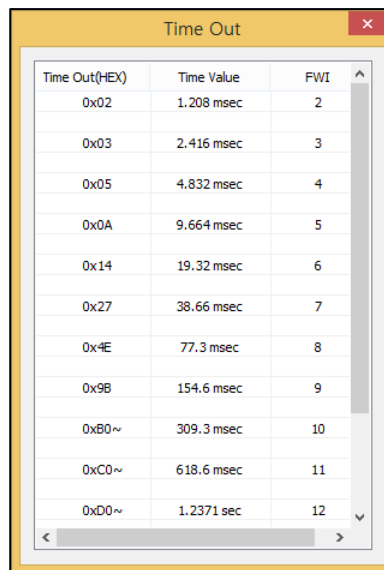
- : Select the communication mode of the Target.



- : Set the waiting time between sending commands and getting a response.

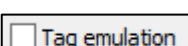


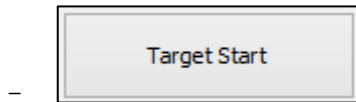
- : Click to get more information. The Time Out box, <Figure 10-4>, will appear.



Time Out(HEX)	Time Value	FWI
0x02	1.208 msec	2
0x03	2.416 msec	3
0x05	4.832 msec	4
0x0A	9.664 msec	5
0x14	19.32 msec	6
0x27	38.66 msec	7
0x4E	77.3 msec	8
0x9B	154.6 msec	9
0xB0~	309.3 msec	10
0xC0~	618.6 msec	11
0xD0~	1.2371 sec	12

<Figure 10-4>
Time Out

-  : Check the box to use the SNEP mode
-  : Check the box to use the Tag Emulation mode



DualCardDll API :

```
DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]);
LLC_Send_Text([PORT],[MODE],[SLEN],[SBUF],[RLEN],[RBUF]);
DE_NFC_Init_TARGET([PORT],[TIMEOUT],[MODE],[RLEN],[RBUF]);
DE_NFC_GetTargetData([PORT],[RLEN],[RBUF]);
DE_NFC_SetTargetData([PORT],[SLEN],[SBUF]);
DE_NFC_GetTargetData([PORT],[RLEN],[RBUF]);
```

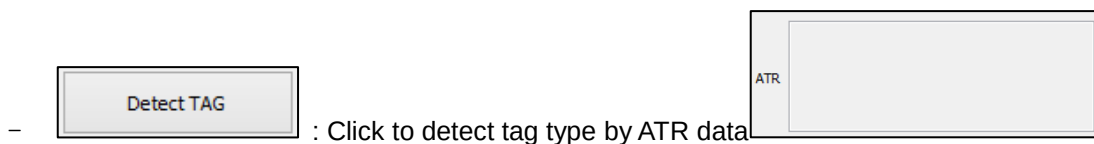
DUALi RW Protocol :

```
BB[TAG TYPE][DATA]
BC[MODE=20][DATA]
9574[MODE]00[TIMEOUT]
96[DATA]
```

10.3 NFC Tag



<Figure 10-5>
Detect NFC Tag



- : Click to detect tag type by ATR data

DualCardDll API :

```
DE_ByPassCommand(m_pDoc->m_nPort, nSlen, 0x83, SBUF, &outlen, RBUF);
```

DUALi RW Protocol :

83

10.4 Message Transmission

SEND ☒ ASCII	DUALi Inc. is an innovative smartcard solution provider, specialized in secured high-end solutions, such as e-ID, e-Passport, e-Payment & NFC.
RECV ☒ ASCII	

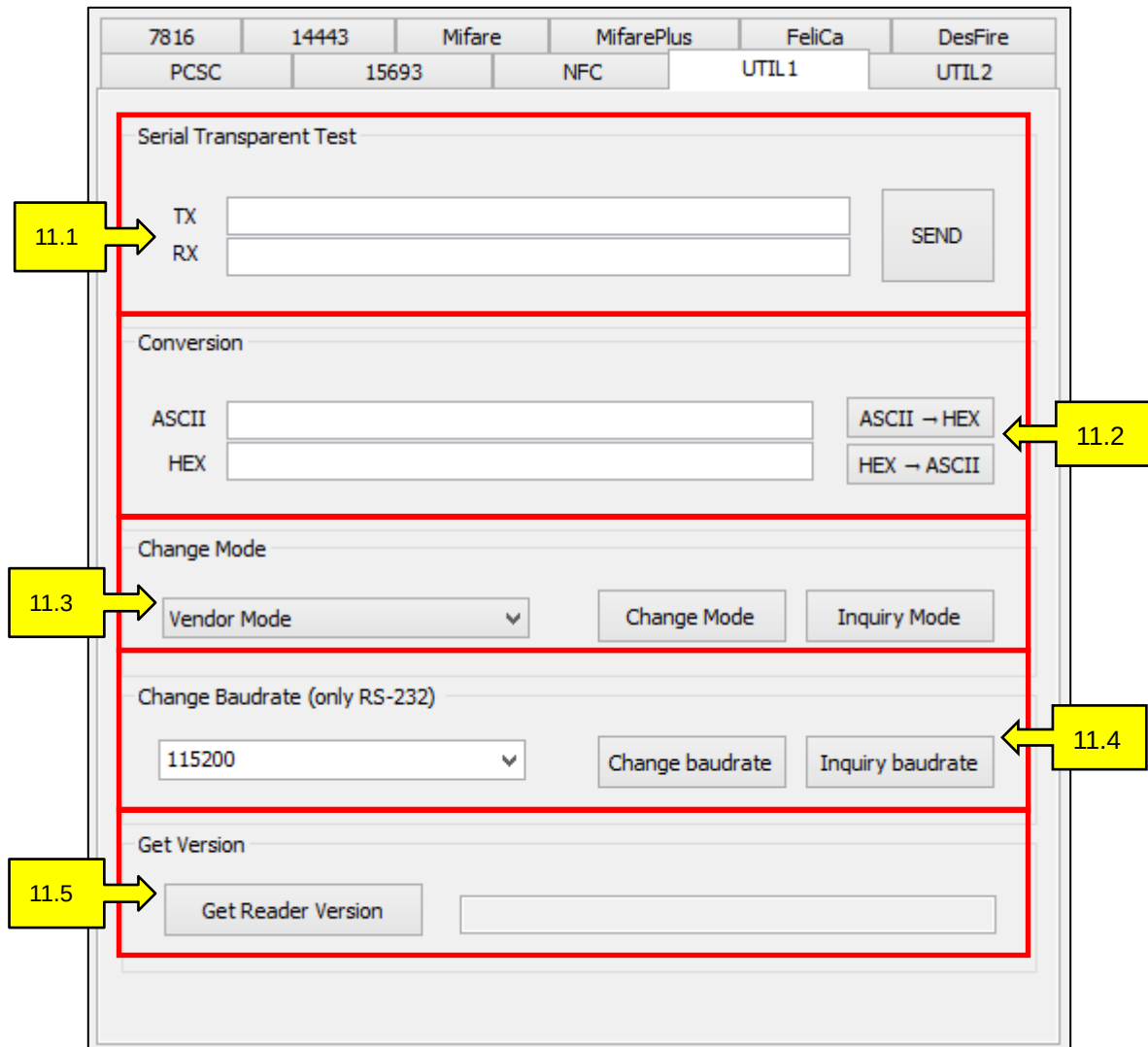
<Figure 10-6>

NFC Message Transmission

Use this group box to transmit messages, using ASCII or Hexa mode, to and from NFC devices.

11. UTIL1

This tab, <Figure 11-1>, has various functions for the user's convenience.



<Figure 11-1>

UTIL1 Tab

11.1 Serial Transparent Test



The interface is titled "Serial Transparent Test". It contains two text input fields labeled "TX" and "RX". To the right of these fields is a blue button labeled "SEND".

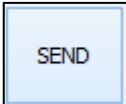
<Figure 11-2>

Serial Transparent Test

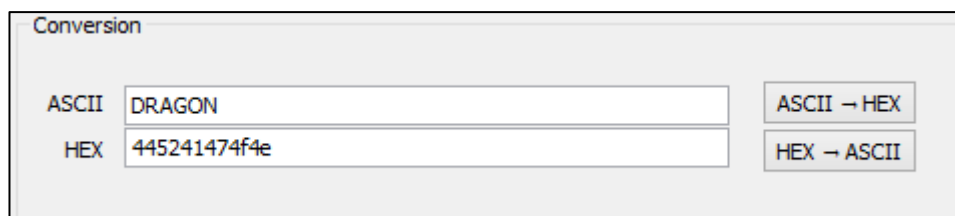
- This group box tests the sending and receiving of data via UART.

- : Input the command/data to send.

- : The response data from the serial equipment is displayed.

- : Click to send input data.

11.2 Conversion Data Type

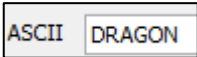


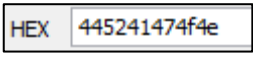
The interface is titled "Conversion". It has two text input fields. The first is labeled "ASCII" and contains the text "DRAGON". The second is labeled "HEX" and contains the text "445241474f4e". To the right of these fields are two buttons: "ASCII → HEX" and "HEX → ASCII".

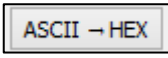
<Figure 11-3>

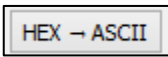
UTIL1 Conversion

- This function is used to convert from a ASCII value to a HEX value or from a HEX value to a ASCII value.

- : Input the ASCII value

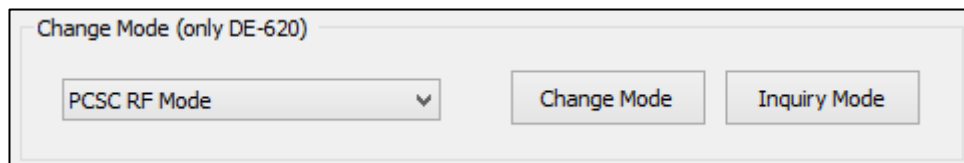
- : Input the HEX value

- : Click to convert from a ASCII value to a HEX value

- : Click to convert from a HEX value to a ASCII value.

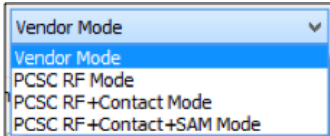
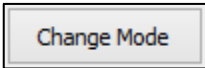
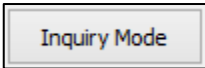
11.3 Change Mode

Users can easily change from Vendor Mode to PCSC Mode, or vice versa, when using USB connected devices.

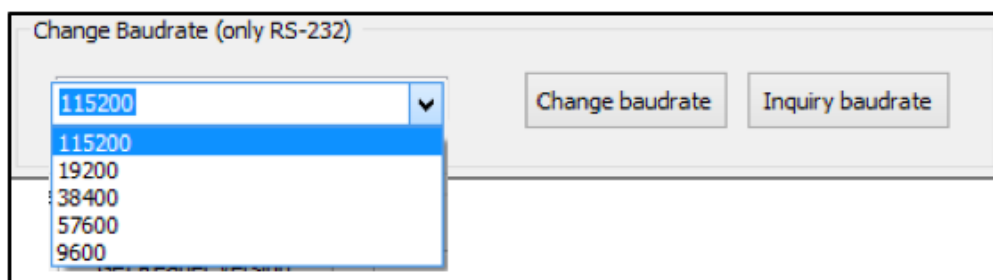


<Figure 11-4>

Change Modes

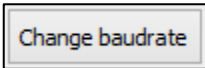
- : Select the mode to change to
- : Click to execute the change
- If the PCSC driver is not yet installed, you have to install the driver after you change to the PCSC mode.
- **DualCardDll API** : `DE_ChangeDevice([SBUF],[MODE],[INQ FLAG]);`
- **DUALi RW Protocol** : `1501[MODE]01`
- : Click to view the current mode of the connected device

11.4 Change Baudrate



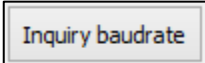
<Figure 11-5>

Change Baudrate

-  :

DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]);

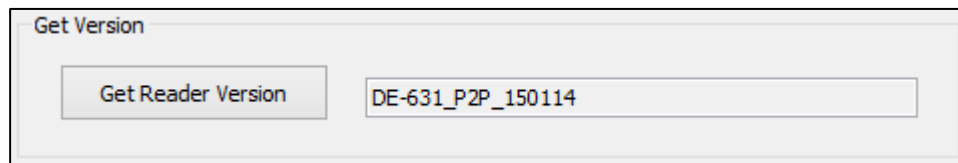
DUALi RW Protocol : 1502[BAUD]

–  :

DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]);

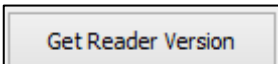
DUALi RW Protocol : 1503

11.5 Get Version



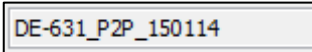
<Figure 11-6>

Reader Version

–  : Click to get the reader's version.

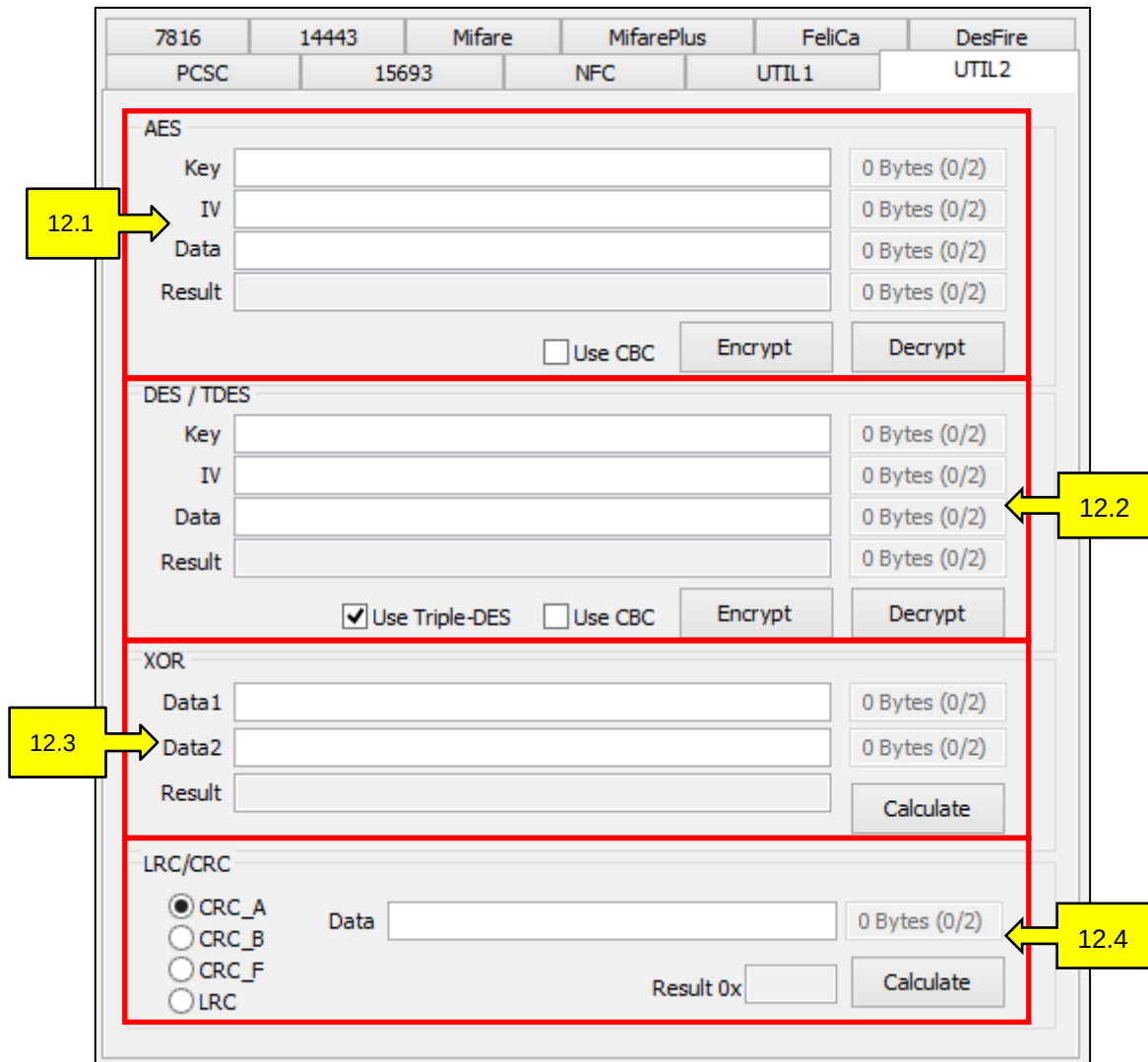
DualCardDll API : DE_GetVersion([PORT],[RLEN],[RBUF]);

DUALi RW Protocol : 16

–  : The reader's version will be displayed in the field.

12. UTIL2

By using DualCard, users can check encrypted and decrypted values of AES, DES, and TDES algorithms. In this tab, there are XOR, LRC, and CRC calculation functions for the user's convenience.



The screenshot shows the UTIL2 tab in the DualCard software. The interface is organized into four main sections, each with a red border and a corresponding callout box:

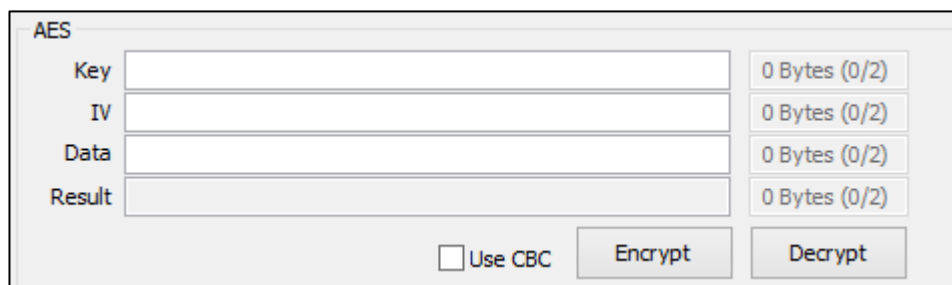
- AES (Callout 12.1):** Contains input fields for Key, IV, Data, and Result, each with a "0 Bytes (0/2)" indicator. It also has a "Use CBC" checkbox and "Encrypt" and "Decrypt" buttons.
- DES / TDES (Callout 12.2):** Contains input fields for Key, IV, Data, and Result, each with a "0 Bytes (0/2)" indicator. It has checkboxes for "Use Triple-DES" (checked) and "Use CBC", and "Encrypt" and "Decrypt" buttons.
- XOR (Callout 12.3):** Contains input fields for Data1, Data2, and Result, each with a "0 Bytes (0/2)" indicator. It has a "Calculate" button.
- LRC/CRC (Callout 12.4):** Contains radio buttons for CRC_A (selected), CRC_B, CRC_F, and LRC. It has a "Data" input field with a "0 Bytes (0/2)" indicator, a "Result 0x" field, and a "Calculate" button.

<Figure 12-1>

UTIL2 Tab

12.1 AES

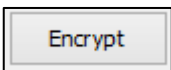
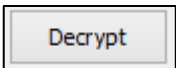
This function is used to check encrypted and decrypted data results of an AES algorithm when using a certain key. Use 16 bytes units.



The AES Algorithm interface is a web-based tool for performing AES encryption and decryption. It features a title bar 'AES' and four input fields: 'Key', 'IV', 'Data', and 'Result'. Each field has a corresponding '0 Bytes (0/2)' indicator. Below the input fields, there is a checkbox labeled 'Use CBC' and two buttons: 'Encrypt' and 'Decrypt'.

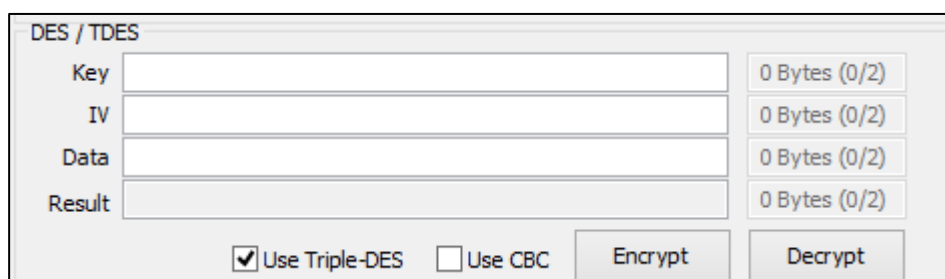
<Figure 12-2>

AES Algorithm

- : Input Key.
- : Input Initial Vector in CBC mode.
- : Input data to encrypt and decrypt. To encrypt data, input plain data. To decrypt data, input encrypted data.
- : The encrypted and decrypted results will be displayed.
- : Check the box to use the CBC mode. When unchecked, the mode is ECB.
- : Click to encrypt the input data.
- : Click to decrypt the input data.

12.2 DES / TDES

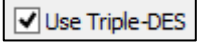
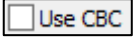
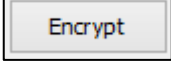
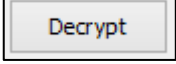
This function is to check the results of data after the TDES algorithm. DES operates with 16 byte units and TDES operates with 8 byte units.



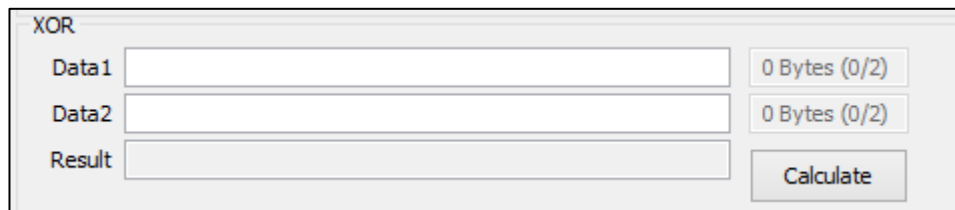
The DES / TDES Algorithms interface is a web-based tool for performing DES and TDES encryption and decryption. It features a title bar 'DES / TDES' and four input fields: 'Key', 'IV', 'Data', and 'Result'. Each field has a corresponding '0 Bytes (0/2)' indicator. Below the input fields, there are two checkboxes: 'Use Triple-DES' (checked) and 'Use CBC' (unchecked), and two buttons: 'Encrypt' and 'Decrypt'.

<Figure 12-3>

DES / TDES Algorithms

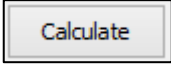
- : Input Key.
- : Input initial vector in CBC mode.
- : Input data to encrypt and decrypt. To encrypt data, input plain data. To decrypt data, input encrypted data.
- : The encrypted and decrypted results will be displayed.
- : Check the box to run TDES. If the box is not checked, the system will run DES.
- : Check the box to use the CBC mode. If the box is not checked the system will use the ECB mode.
- : Click to encrypt plain data. Results will populate.
- : Click to decrypt encrypted data. Results will populate.

12.3 Exclusive OR



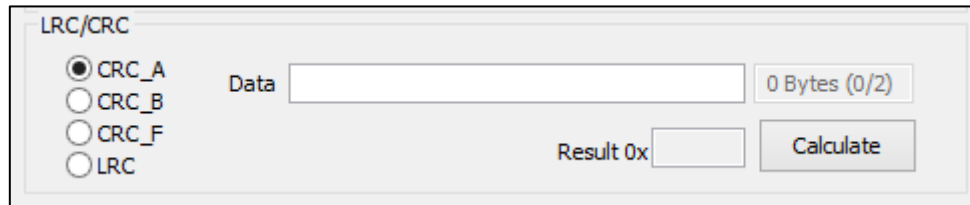
The interface is titled "XOR". It contains three input fields: "Data1", "Data2", and "Result". To the right of "Data1" and "Data2" are text indicators showing "0 Bytes (0/2)". To the right of the "Result" field is a "Calculate" button.

<Figure 12-4>
Exclusive OR

- To calculate the XOR of either Data1 or Data2, double click either of the two empty fields.
- : Click to calculate XOR. Results will display in the "Result" field.

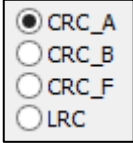
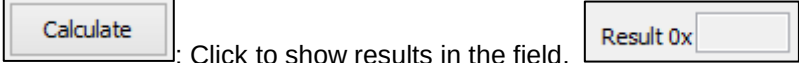
12.4 LRC / CRC

The LRC / CRC function is used to check for redundancy in send/receive data.



The screenshot shows a software interface titled "LRC/CRC". It contains four radio buttons for selecting a mode: "CRC_A" (selected), "CRC_B", "CRC_F", and "LRC". To the right of these buttons is a "Data" input field followed by a status indicator "0 Bytes (0/2)". Further right is a "Result 0x" input field and a "Calculate" button.

<Figure 12-5>
LRC/CRC

- : Select mode to check for redundancy.
- : Insert data
- : Click to show results in the field,

13. Annex

13.1 Mifare Commands

Command	Explanation
Auto Read	Read card automatically DualCardDII API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]) DUALi RW Protocol : 3600[AUTH MODE][BLOCK NO][KEY]; 27[BLOCK NO]
Auto write	Write card automatically DualCardDII API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]) DUALi RW Protocol : 3800[AUTH MODE][BLOCK NO][KEY][DATA] 28[BLOCK NO][DATA]
Decrement Transfer	Decrement the value and transfer to selected block DualCardDII API : DEA_Dec_Transfer([PORT],[BLOCK NO],[VALUE],[TRANSFER BLOCK NO]); DUALi RW Protocol : 2C[BLOCK NO][VALUE][TRANSFER BLOCK NO]
Decrement Transfer, Verify by Reader	Decrement to transfer and verify by reader DualCardDII API : DEA_Dec_Transfer2([PORT],[BLOCK NO],[VALUE],[TRANSFER BLOCK NO]); DUALi RW Protocol : 34[BLOCK NO][VALUE][TRANSFER BLOCK NO]
Increment Transfer	Increment to transfer. DualCardDII API : DEA_Inc_Transfer([PORT],[BLOCK NO],[VALUE],[TRANSFER BLOCK NO]); DUALi RW Protocol : 2B[BLOCK NO][VALUE][TRANSFER BLOCK NO]
Increment Transfer, Verify by Reader	Increment to transfer and verify by reader. DualCardDII API : DEA_Inc_Transfer2([PORT],[BLOCK NO],[VALUE],[TRANSFER BLOCK NO]); DUALi RW Protocol : 33[BLOCK NO][VALUE][TRANSFER BLOCK NO]
Loadkey	Load key data to reader DualCardDII API : DEA_Loadkey([PORT],[MODE],[KEY NO],[KEY]); DUALi RW Protocol : 2F[MODE][KEY NO][KEY]
Request Auth	Request, Anticollision, Select, Authentication with KEY No DualCardDII API : DEA_Req_Auth([PORT],[REQUEST MODE],[AUTH MODE],[KEY NO],[BLOCK NO],[RLEN],[RBUF]); DUALi RW Protocol : 3100[AUTH MODE][KEY NO][BLOCK NO]
Request Authkey	Request, Anticollision, Select, Authentication with KEY ffffffff DualCardDII API : DEA_Req_Authkey([PORT],[REQUEST MODE],[AUTH MODE],

	[BLOCK NO],[KEY],[RLEN],[RBUF]); DUALi RW Protocol : 3200[AUTH MODE][BLOCK NO][KEY]
Request to Read with Key No	Request to read with KEY No DualCardDll API : DEA_Req_AuthRead([PORT],[REQUEST MODE],[AUTH MODE],[KEY NO],[BLOCK NO],[RLEN],[RBUF]); DUALi RW Protocol : 3500[AUTH MODE][KEY NO][BLOCK NO]
Request to Read with KeyData	Request to read with KEY DualCardDll API : DEA_Req_AuthkeyRead([PORT],[REQUEST MODE],[AUTH MODE],[BLOCK NO],[KEY],[RLEN],[RBUF]); DUALi RW Protocol : 3600[AUTH MODE][BLOCK NO][KEY]
Request to Select	Request to select DualCardDll API : DEA_Req_Select([PORT],[REQUEST MODE],[RLEN],[RBUF]); DUALi RW Protocol : 3900
Request to Write with Key No	Request to write with KEY No DualCardDll API : DEA_Req_AuthWrite([PORT],[REQUEST MODE],[AUTH MODE],[KEY NO],[BLOCK NO],[DATA],[RLEN],[RBUF]); DUALi RW Protocol : 3700[AUTH MODE][KEY NO][BLOCK NO][DATA]
Request to Write with KeyData	Request to write with KEY DualCardDll API : DEA_Req_AuthkeyWrite([PORT],[REQUEST MODE],[AUTH MODE],[BLOCK NO],[KEY],[DATA],[RLEN],[RBUF]); DUALi RW Protocol : 3800[AUTH MODE][BLOCK NO][KEY][DATA]
Write for Mifare Ultra Light	Write command for Mifare Ultra Light DualCardDll API : DEA_UltraM_Write([PORT],[ADDRESS],[DATA]); DUALi RW Protocol : 3B[ADDRESS][DATA]
Req Idle	Request idle command DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]) DUALi RW Protocol : 21
Write and Read, Verify by App	Write and read Mifare and verify by application DualCardDll API : SecReqIdle OnBnClickedButtonAnticollision OnBnClickedButtonSelect SecAuthKey SecWrite SecRead DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]) DUALi RW Protocol :
Write and Read, Verify by Reader	Write and read Mifare and verify by reader DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]) DUALi RW Protocol : AA[FROM BLOCK NO][TO BLOCK NO][KEY][DATA]

13.2 FeliCa Commands

Command	Explanation
Polling	Get a value of IDm and PMm
13.2 Polling-No Auth with RC-251	Get a value of IDm and PMm without RC-251 Auth
Mutual	Authentication command for traditional method
13.2 Read Block with ENC	Read command for encrypted Service
13.2 Write Block with ENC	Write command for encrypted Service
Read Block without ENC	Read command for a non-encrypted Service
13.2 Write Block without ENC	Write command for a non-encrypted Service
Read Block without ENC(Token)	Read command for a non-encrypted Service(Token)
13.2 Write Block without ENC(Token)	Write command for a non-encrypted Service(Token)
13.2 Increment with ENC	Increment command for encrypted Service
Decrement with ENC	Decrement command for encrypted Service
13.2 Cashback without ENC	Cashback command with non-encrypted Service
13.2 Read Balance	Read Balance

13.3 DesFire Commands

Command	Explanation
13.2 Create Application	This command allows the user to create new applications on the PICC. You need to set AID : 000001 to Create Application. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020006CA[AID][KEY SET][KEY NO]
Delete Application	This command allows the user to permanently deactivate applications on the PICC. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020004DA[AID]
Get Application IDs	This command returns the Application IDs of all active applications on a PICC. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]);

	DUALi RW Protocol : 3A0200016A
Get Key Setting	This command allows the user to get information on the PICC and application master key settings. In addition, it returns the maximum number of keys that can be stored in the selected application. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A02000145
Change Key Settings	This command changes the master key settings depending on the selected AID. AID : 000001 DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A02000154[KEY SET]
Change Key	This command allows the user to change any key stored in the PICC. You need to input Key No in 0 . DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A02[FLAG]02C4[KEY NO][KEY]
Get Version	DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A02000160
Get Available Free Memory	DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A0200016E
Format Card	This command releases the PICC user memory. DualCardDII API :

	DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020001FC
Commit Transaction	This command allows the user to validate all previous write access on backup data files, value files, and record files in one application. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020001C7
Create File	This command creates files according to the selected file type File Type Standard Data . (Standard Data, Backup Data, Value, Linear Record, Cyclic Record) You need to set one of options File Size 80 , Value 80 Lower Limit 0 Upper Limit 65536 ☐ Limited Credit Enable or Record Size 80 Max Num 2 DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020008CD[FILE NO][COMM SET][ACCESS RIGHT][FILE SIZE] 3A020008CB[FILE NO][COMM SET][ACCESS RIGHT][FILE SIZE] 3A020008CC[FILE NO][COMM SET][ACCESS RIGHT][LOWER LIMIT][UPPER LIMIT][VALUE] 3A020011C0[FILE NO][COMM SET][ACCESS RIGHT][RECORD SIZE][NUM RECORD] 3A020011C1[FILE NO][COMM SET][ACCESS RIGHT][RECORD SIZE][NUM RECORD]
Delete File	This command permanently deactivates a file within the file directory of the selected application. DualCardDII API :

	DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020002DF[FILE NO]
Get File IDs	This command returns the File IDs of all active files in the selected application. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A0200016F
Get File Setting	This command allows the user to get information on the properties of a specific file. The information provided by this command depends on the type of file that is queried. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A0200025F[FILE NO]
Change File Settings	This command changes the access parameters of an existing file. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020005F5[FILE NO][COMM SET][ACCESS RIGHT]
Change File Settings-Free Access	This command changes the access parameters of an existing file for free access. DualCardDII API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN],[SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020005F5[FILE NO][COMM SET]EEEE

14. Example

14.1 Add Commands to the Packet List and Save/Load Command Sets

- Add Commands to the Packet List:

Log

IDX	Data
5	=>3A0100
6	<=0097C22199F879F8D6671F11222BC50D9E556677882C270FF39000
7	=>150100
8	<=00
9	=>83
10	<=003BF99100FF918171404000A80044E2AA^431C8062
11	=>150100
12	<=00
13	=>83
14	<=003BF99100FF918171404000A80044E2AA^431C8062

1. Double click the command

☒ Write Check ☐ Error detection Err Analysis Text File List Clear

Commands Set

0/0 Loop Count 1 Run

IDX	Command	Delay	Comment
☒ 1	150 100	50	
☒ 2	83	50	

2. Selected Command will be added to the Commands Set

Group COS TEST Add Group Packet List Clear

Name Save Load Update DB

3. Input loop count. This number decides the number of loop (Execution of selected Command)

4. Click to execute the command in the Packet List.

- Save Command Set

Name

1. Enter the name of the command set to save

2. Click to save

- Load/Delete Command Set

Name

1. Click to load

LOAD

Group: **VENDOR**

Datetime	Name	
2014/01/06 14:58	device test	[DEL]
2013/11/14 23:03	FeliCa	[DEL]
2013/11/14 23:04	iso 15693	[DEL]
2013/11/14 23:04	Kovio	[DEL]
2013/11/14 23:04	MIFARE	[DEL]
2013/11/14 23:04	MIFARE Batch	[DEL]
2013/11/14 23:04	MIFARE_UL	[DEL]
2013/11/14 22:59		
2014/01/06 20:57		
2013/11/14 22:58		
2014/01/06 20:56		
2014/01/06 21:01	z p2p	[DEL]
2014/01/27 22:11	z p2p Target	[DEL]

2. Double click on command set to open

3. Double Click on [DEL] to delete the selected command set

14.2 Mifare

- [Auth Key] Authentication

REQA

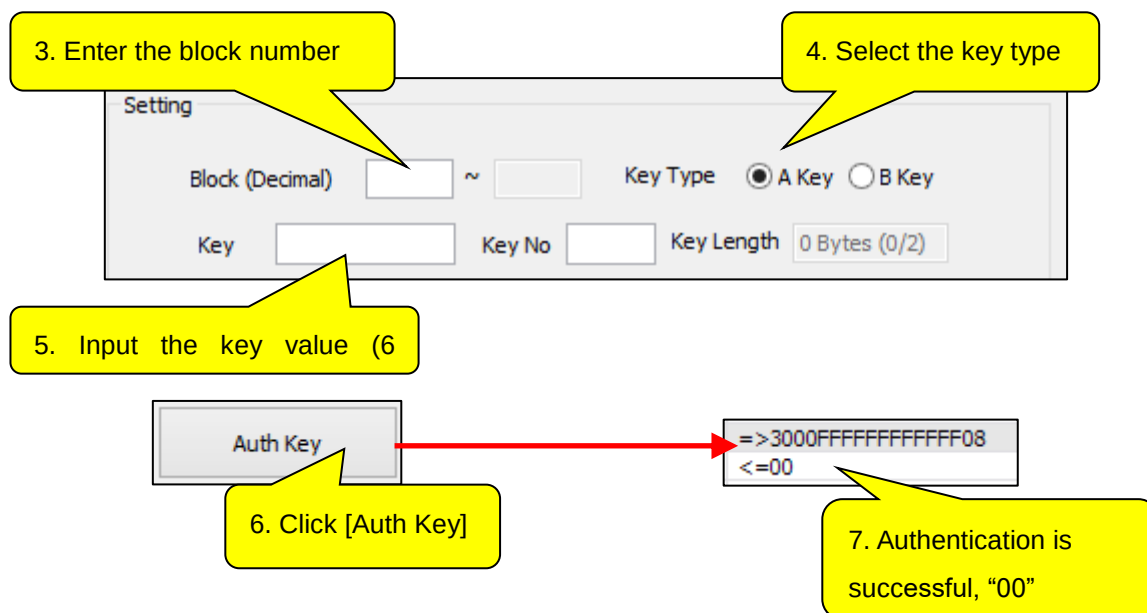
AntiCollision (cascade level)

1. Click [REQA]

2. Click [Anti Collision]

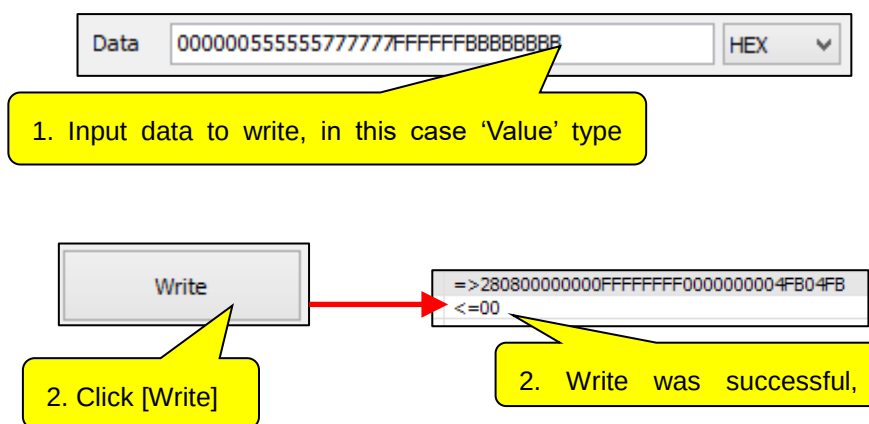
27	=>21
28	<=000400
29	=>3D
30	<=008613AD328

3. The UID of the Mifare card will display. In this case, "613AD328".



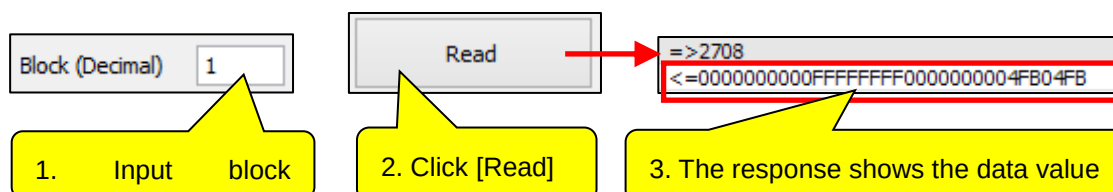
– [Write]

The write process has to be done after the [Auth Key] procedure



– [Read]

The read process has to be done after the [Auth Key] procedure



- [Decrement]

The decrement process has to be executed after the [Auth Key] process. For decrement, the block data must be a Value type.

1. Input block number as a Value type

2. Input Data(amount to deduct) for Decrement (HEX)

3. Click [Decrement]

4. Click [Restore]

5. Click [Transfer]

Transaction list:

- =>2708
- <=0010000000EFFFFFFF1000000004FB04FB
- =>2A0810000000
- <=00
- =>2E08
- <=00
- =>2D08
- <=00
- =>2708
- <=0000000000FFFFFFF0000000004FB04FB

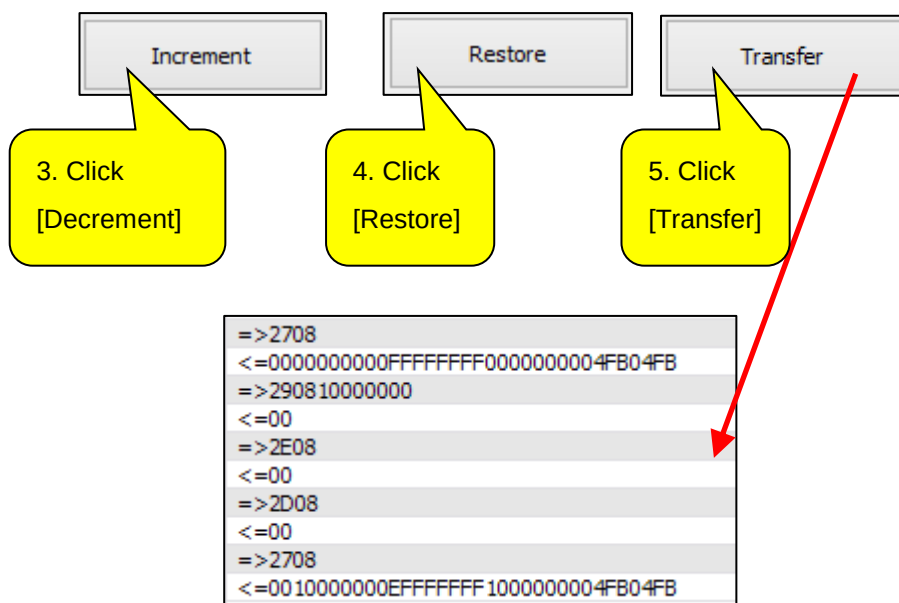
- [Increment]

The Increment process has to be done after the [Auth Key] procedure. For Increment, block data must be a 'Value type'.



The screenshot shows the 'Block (Decimal)' field with the value '8' and the 'Data' field with the value '10000000'. A dropdown menu is set to 'HEX'. Two yellow callout boxes provide instructions:

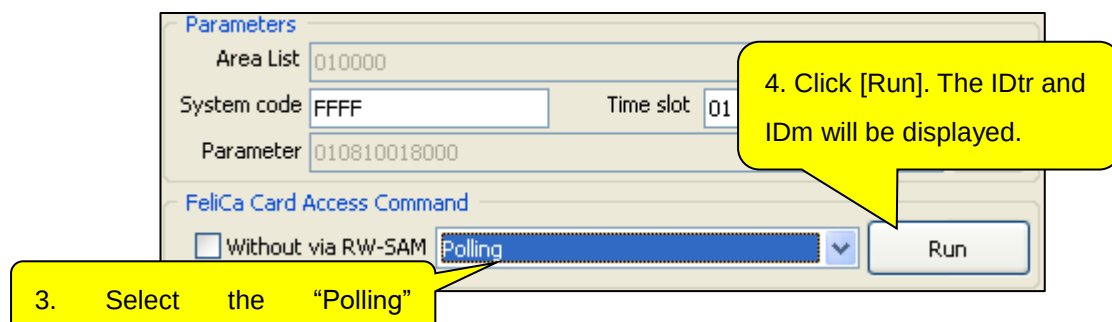
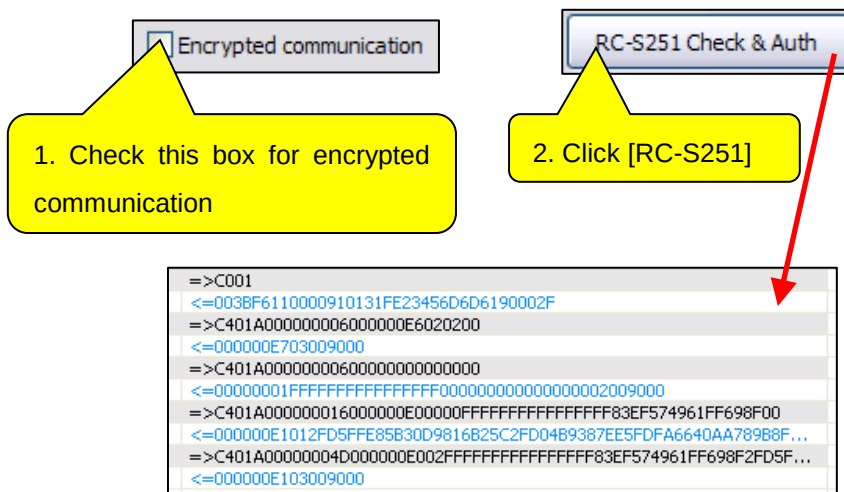
1. Input block number as a Value type
2. Input Data for increment (amount to



14.3 FeliCa

This procedure is based on FeliCa's SAMPLE 5 cards.

- [Read]



IDtr and IDm are displayed

7. Click [Run]. The block data will be displayed in Data box

6. Select “Read Block with

9. Input Area Code,
Service code

10. Input Block List

11. Input Area List

12. Click [Generate List]

A list of values from the Service & Block list is generated

```

=>C401A00000002A000000862900D67E2DDC8E6601010801570CE50F010000...
<=0101C71001010801570CE50F010000010810AEFC5736EE8D952A9000
=>50181001010801570CE50F010000010810AEFC5736EE8D952A90
A1101010801570CE50FC670B5CBF225C4E2FE9F520DFD28793D
A00000001C0100001101010801570CE50FC670B5CBF225C4E2FE9F520...
C71201010801570CE50FE81A3FAF1D88A50A9000
1201010801570CE50FE81A3FAF1D88A50A90
FF8B6E895857B9BAE62CA99AE7D477D0A52F295713E89ED2C805...
A0000000240100001307FF8B6E895857B9BAE62CA99AE7D477D0A52F...
00000872A00D67E2DDC8E6600000102030405060708090A0B0C0D0E0F00...
=>C401A000000011000000882B00D67E2DDC8E66000001800000
<=01022E1497278CC66B4545FB8000A94AC9A64E759B142A2D17ABA6369000
=>501A1497278CC66B4545FB8000A94AC9A64E759B142A2D17ABA63690
<=002A1586DAC50C041C8757015617019F303029C999FFE8617DEF6B9A59C...
=>C401A00000002C0100001586DAC50C041C8757015617019F303029C999F...
<=000000892C00D67E2DDC8E66000001123410081008100810081008100...

```

Value of selected block

- [Write]

The RW-SAM authentication process is required before the Write process.

Data

Block No. for RC-S890

Data Length

Data

1. Input data to write

FeliCa Card Access Command

☐ Without via RW-SAM

2. Select "Write Block with

3. Click [Run]

```

=>C401A00000002A000000864500D67E2DDC8E6601010801570CE50F010000...
<=0101C71001010801570CE50F010000010810F3685543E338C6ED9000
=>50181001010801570CE50F010000010810F3685543E338C6ED90
<=001A1101010801570CE50F018D3D73FEEAA2D9E321A1B2FFBA92A3
=>C401A00000001C0100001101010801570CE50F018D3D73FEEAA2D9E321A1...
<=0101C71201010801570CE50FF75DB30D8B09A9179000
=>50121201010801570CE50FF75DB30D8B09A91790
<=0022131CF2D9614D0424E48E5B8657195E69DE3C17BE1AAEC989C262F6E...
=>C401A000000024010000131CF2D9614D0424E48E5B8657195E69DE3C17BE1...
<=000000874600D67E2DDC8E6600000102030405060708090A0B0C0D0E0F00...
=>C401A0000000210000008A4700D67E2DDC8E660000018000444444444444...
<=0101FD16B770E3E2D5395B812FA099BCE9FC3CB13DC6602B34D53962B807...
=>502A16B770E3E2D5395B812FA099BCE9FC3CB13DC6602B34D53962B8077C...
<=001A17A2E99EEC32D4191C681AB5F8E109FD726FD7C55F5FEE6E00
=>C401A00000001C01000017A2E99EEC32D4191C681AB5F8E109FD726FD7C5...
<=0000008B4800D67E2DDC8E66000009000

```

14.4 DesFire

14.4.1 Basic Operation

This procedure is based on the default key value and standard data file type. AID is '000001'

- [Auth]

The screenshot shows the 'Basic Operation' screen with the following elements and numbered callouts:

- 1. Click [Detect Card]**: Points to the 'Detect Card' button.
- 2. Input AID**: Points to the 'AID' input field, which contains '000001'.
- 3. Click [Select Application]**: Points to the 'Select Application' button.
- 4. Input**: Points to the 'Key' input field, which contains a long string of zeros.
- 5. Click [Authenticate]**: Points to the 'Authenticate' button.

- [Standard Data Read]



The screenshot shows the 'Standard Data' dialog box with the following elements and callouts:

- 1. Select [File]**: Points to the 'File No' dropdown menu.
- 2. Input [Offset]**: Points to the 'Offset' text input field.
- 3. Input [Size]**: Points to the 'Size' text input field.
- 4. Click [Read]**: Points to the 'Read' button.

The dialog box contains a 'Standard Data' dropdown, 'File No' dropdown, 'Offset' and 'Size' text inputs, 'Write', 'Read', and 'Debit' buttons, and a 'Standard Data' label.

[illegible]

- [Standard Data Write]

Standard Data Write

1. Select [File]

Standard Data File No 0 Offset 0 Size 8

Write Read

2. Input [Offset]

3. Input [Size]

5. Click [Write]

4. Input [Data]

Data 0123456789ABCDEF Bytes (2/2)

[illegible]

- [Change Master Key]

The Key '0' Authentication process is required before the Change Master Key process.

Command List **Change Key**  

1. Select "Change Key" in command

2. Click [Run]

Setting

AID

Key No

Key

Key Set 0x ...

Key Type ☒ DES/3DES ☐

Run

3. Select key number

4. Input new key

5. Run

16 Bytes (2/2)

3. Select key number

4. Input new key

5. Click [Run]

[illegible]

“Change Key” Response

Authenticate with new key

14.4.2 Batch Command

- [Batch Read]

The screenshot shows the 'Batch Function' dialog box with the following fields and buttons:

- AID:** 000001
- Key No:** 0 (dropdown)
- 24 Bytes (2/2)** (text)
- Key:** 00
- Set Parameter** (button)
- ☐ **Data Encryption**
- File No:** 0 (dropdown)
- Offset:** 0
- Size:** 8
- Value:** 16
- Batch Write** (button)
- Batch Read** (button)
- Batch Credit** (button)

Numbered callouts indicate the following steps:

1. Input key data (points to the Key field)
2. Click [Set Parameter] (points to the Set Parameter button)
3. Input file data (points to the File No field)
4. Click [Batch Read] (points to the Batch Read button)

1. Input key data

2. Click [Set Parameter]

4. Click [Batch Read]

3. Input file data

[illegible]

“Batch Read” Response

- [Batch Credit]

[illegible]